

Descending Pathways to the Spinal Cord, III: Sites of Origin of the Corticospinal Tract

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ABSTRACT

The somata of corticospinal neurons were labeled with horseradish peroxidase that had been applied to a hemisection of the spinal cord at the C1-C2 junction in 22 species of mammals. After tetramethylbenzidine processing, with and without counterstaining with cresyl violet or neutral red, the labeled cells in systematic sets of sections throughout the cerebral cortex were plotted and counted. Several morphological features of the corticospinal cells were examined including their cell type, number, density, concentration, laminar distribution, and their distribution across the cortical surface.

The results show that the labeled corticospinal neurons were invariably layer V pyramidal cells. However, in many mammals they were found to be stacked one above the other within layer V, sometimes many neurons deep. Despite the concentration of corticospinal neurons within layer V, many unlabeled neurons were also present within the layer throughout the extent of the labeled region.

The results also indicate that at least two spatially distinct regions of neocortex originate corticospinal fibers in each of the animals in the sample. In addition to these two regions, a third segregated region is present in the cortex of primates and an apparently different third region is present in the cortex of Glires (Rodentia and Lagomorpha). The third region of corticospinal cortex in primates is located on the lateral surface of the cortex in prosimians and New World monkeys and is buried in the caudal bank of the inferior arcuate sulcus in Old World monkeys.

The results also show a predominantly contralateral corticospinal tract in all but 4 of the 22 mammals in the sample. Although these 4 mammals are each members of the order Insectivora, a less modified member of the same order possessed the predominantly contralateral projection of most mammals, hence denying the notion that a predominantly ipsilateral tract is a characteristic of Insectivora.

Key words: cortex, motor system, evolution

At least since 1853, when Cruveilhier discovered a correlation between cerebral atrophy, shrinkage of the medullary pyramids, and contralateral paralysis, the corticospinal tract (or CST) has held a favored position in clinical and experimental neurology (see Lassek, '54). The functional implications of a large and direct pathway from human cerebral cortex to spinal motor neurons have not escaped the attention of each intervening generation of students of the brain. However, despite the large number of experimental and clinical reports on the CST and despite the availability of a long succession of anatomical tract-tracing methods, the cerebral locations originating the tract have still not been outlined in their entirety.

In a series of studies of the systems descending to the spinal cord, we have collected comparative data on the cortical sites originating the corticospinal tract (Nudo and Masterton, '88). In these studies to date, each of 103

individual mammals in 24 species was treated with exactly the same procedure—hemisecting the spinal cord at the C1 and C2 junction and placing raw HRP on the cut fibers. This procedure resulted in the heavy retrograde labeling of vast numbers of neural somata distributed throughout the neuraxis from coccygeal spinal cord to cerebral neocortex (Nudo and Masterton, '88). It is this body of materials that provides the data on the corticospinal tract presented here.

In the present report, the gross similarities and variations in distribution of the cortical cells originating the CST are described in several diverse groups of mammals. Some quantifiable details will be described in subsequent papers (Nudo and Masterton, '90). Preliminary reports of some of

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these data have been presented in abstract form (Nudo and Masterton, '85, '86).

MATERIALS AND METHODS

The selection of a retrograde tract-tracing procedure that would maximize the labeling of neocortical cell groups, the experimental basis for the surgical and histological techniques eventually selected, the selection of the species to be included, and the screening of the histological materials prior to analysis have been described in detail in previous reports (Nudo and Masterton, '88, '89). Briefly, we conducted a series of preliminary experiments with rats by using several retrograde tracers, sites, and methods of application, survival times, and chromagens. Because in our hands, the HRP bulk-filling procedure of Hicks and D'Amato ('77) was not surpassed by any other technique for labeling the largest numbers of supraspinal neurons in each species, it was adopted as a standardized procedure for the entire series of studies (also see Toyoshima and Sakai, '82).

In our version of this procedure, flakes of raw HRP were applied directly to the freshly cut fibers of the spinal cord after it had been hemisected at the C1-C2 junction under deep Chloropent anesthesia. After a 3 day survival period, the animals were deeply anesthetized once more and after intracardial perfusion and fixation, the resulting brain materials were sectioned and then processed with tetramethyl benzidine (Mesulam, '78). This procedure was directly applicable to each species and therefore, each animal was treated in a virtually identical way. However, the individual cases were admitted for reconstruction and eventually for quantification only after a rigorous screening of the histological materials. The screening procedure served to eliminate those with either incomplete or misplaced hemisections, incomplete exposures of the cut fibers to HRP, extracellular HRP in the dorsal column nuclei, or cell labeling suggesting HRP transport via blood or cerebrospinal fluid (Nudo and Masterton, '88).

Species represented

Of the total of 103 individual adult animals, the histological materials from 56 individuals representing 22 species in 20 genera survived the histological screening process and are reported here. These 56 individuals proved to have a complete hemisection and an exposure to the HRP to or beyond the cord's midline. Moreover, the dorsal column nuclei were free of extracellular HRP and there was no evidence whatever of HRP transport via blood or CSF. These 22 species in 20 genera, in turn, represent 14 families, 7 orders, and 2 subclasses of Mammalia. This list includes each of the species reported previously (Nudo and Masterton, '88, '89) plus one other species: the least shrew (*Cryptotis parva*). It does not include 26 individuals eliminated because of incomplete hemisections including 6 individuals representing 2 entire species (rock hyrax, armadillo), nor 21 other individuals eliminated because HRP might have invaded the dorsal column nuclei. However, none of these rejected cases contraindicates any result or conclusion derived from the cases presented here, and one species, rock hyrax, will be referred to below despite its incomplete hemisection.

Methods of reconstruction and analysis

In order to reconstruct the distribution of retrogradely labeled neurons in cerebral cortex, systematic sets of TMB-

processed, non-counterstained, 50 μm sections were examined by use of light microscopy. A sample consisting of every tenth section through the cerebral cortex in the 5 species with brains weighing more than 25 g and every fifth section in the 12 species with brains weighing less than 25 g was used for analysis. For the 5 species with brains weighing less than 1 g, a second or third set of every fifth section was added. In these series of sections, each labeled cell-profile was counted and its location plotted on an enlarged drawing of the section with the aid of an x-y pantagraph slaved to the stage of a light microscope.

The surface distribution of HRP-labeled neurons across the cortical surface was then reconstructed from the plots both in dorsal and lateral perspectives. Two forms of reconstruction proved useful: *first*, a two-dimensional, radial reconstruction that simply located each labeled cell on the overlying cortical surface (e.g., see Murray and Coulter, '81), and *second*, a cell-density plot that showed the relative density of labeled cells per unit area of cortical surface. These density plots also served to provide criteria for distinguishing the cortical regions with particularly high or low densities of labeled neurons. Both types of reconstruction are presented below.

Cell-density distribution. In order to derive the density plots, the cell densities per surface area of overlying cortex were estimated by counting the number of HRP-labeled soma profiles subjacent to each 250 μm $\times$ 50 μm area (1/80 mm^2) of neocortical surface within each section of the 1 in 5 section sample.

Because the labeled somata were usually filled completely with HRP reaction product, no intracellular element was visible for the strict stereological estimation of the numbers of labeled cells by usual means (Sterio, '84; Pakkenburg and Gundersen, '88). Instead, profiles of the neuron somata themselves were used as the counting elements and because of their large size relative to section thickness (i.e., 11 to 34 μm somata, 50 μm sections), these raw counts required correction for the resulting overestimation (e.g., Konigsmark, '70; Weibel, '79).

Specifically, the raw profile counts were first converted to an estimate of neuron number by multiplying the profile counts by the number of uncounted intervening sections (i.e., by 5 in most brains, by 10 in large brains) and then corrected for "over-counting" by multiplying with the term: $C = t/[t + d(1 - e)]$; where t = section thickness; d = average soma size; and e = the smallest fragment of a neuron counted as a "labeled profile" (which proved to be about 15% of the average soma size, d). For each animal then, the correction term varied only with the average size of its labeled somata, d : $C = 50/[50 + (0.85 * d)]$. To estimate the average size of the labeled somata (d) in turn, several samples each containing at least 10 labeled somata with apical dendrites visible within the section were viewed by a video camera, magnified, and projected on a computer monitor. Their perimeters were then traced with an electronic drawing pad and their diameters calculated and averaged as if the cells were perfect spheres. These approximations biased the estimate toward slightly lesser numbers of cells—with a 10% error in the estimate of somata diameter resulting in a bias of less than 7% in the estimate of labeled cell number.

For the density maps, the locations of the crest and trough of each sulcus were also plotted so that "unfolded" views of the cortical surface could be drawn (i.e., with the banks and depths of sulci and fissures exposed, see Fig. 2). With the aid

of a microcomputer, the densities of labeled cells could be displayed on a drawing of the unfolded cortical surface in any of a number of forms or levels of resolution such as absolute numbers of cells per one or another unit area of cortical surface, percentages of maximum density, normalized densities, etc. Later, these various forms of cell-density maps allowed derivation of quantitative criteria for identifying the segregated regions or clusters of labeled cells across the cortical surface.

Identification of cytoarchitectonic areas. In addition to the set of sections that were HRP-processed without counterstain, another set of sections was HRP-processed and Nissl-counterstained in each individual, while still another set was HRP-processed and counterstained with neutral-red. From these materials it was possible in most cases to identify the cytoarchitectonic areas containing the HRP-labeled cells. Prior descriptions of the cortical cytoarchitecture of several of the species served as useful guides to identification of major cytoarchitectonic areas containing labeled neurons (e.g., Gray, '24; Peele, '42; von Bonin, '45; von Bonin and Bailey, '47; Hassler and Muhs-Clement, '64; Kangagasumtheram et al., '66; Sanides and Krishnamurti, '67; Walsh and Ebner, '70; Zilles et al., '78; Fleischhauer et al., '80; Donoghue and Wise, '82).

However, because of the lack of clear-cut structural differentiation in the neocortex of the more generalized mammals (e.g., the opossums or the hedgehogs), the relationship of the distribution of labeled neurons to cytoarchitectonically defined cortical areas could be described only in general outline. For example, in the materials from each of the primates and carnivores, a clear distinction could be made between Area 4 and Areas 3, 1, and 2 on the basis of the relative granularity of layer IV and the presence or absence of large pyramidal cells in layer V. However, for the opossums and hedgehogs, few clear cytoarchitectonic boundaries could be seen across the rostrocaudal extent of either the dorsal or lateral cortex. Therefore, a detailed account of the relationship between the distribution of corticospinal neurons and the cortical cytoarchitecture in these species is best left to more detailed reports of the individual species (e.g., see Martin and Fisher, '68; Rapisarda et al., '85; Miller, '87). Nevertheless, wherever possible, the distributions of labeled neurons were related both to cytoarchitectonic areas and physiologically defined fields and this correspondence is presented below as each group of labeled neurons is described.

RESULTS

Quality of retrograde labeling

In each individual prepared by the procedure adopted as the standard for this study, very large numbers of neurons were vividly labeled at each level of the neuraxis from coccygeal spinal cord through cerebral neocortex. As described previously, at least 28 supraspinal cell groups containing labeled cells could be identified in every species (Nudo and Masterton, '88). The number of labeled neurons in each of these cell groups was far larger than has been reported previously for the same species. We will return to this point again.

The constancy of labeling in the *noncortical* cell groups, as well as the completeness of retrograde filling of individual neural somata within them and within neocortex itself, served as within-individual and within-species controls for the efficacy and relative completeness of the labeling procedure

(Fig. 1). The qualitative constancy within families lent further assurance that no individual-specific, species-specific, or genus-specific HRP-phobia was blocking or seriously attenuating the HRP transport to cortex (Nudo and Masterton, '88).

Labeling of cortical neurons

Turning specifically to neocortex, a large number of labeled neurons were seen contralateral and, to a varying degree, ipsilateral to the HRP application in each case. The labeled somata were invariably pyramidal or some variation of pyramidal (tear-drop, triangular, etc.) which might be expected by changes in the relationship between the constant plane of section relative to the gradually curving cortical surface.

The HRP reaction-product in these labeled neurons typically filled the entire soma, obscuring the nucleus and extending into many if not all of the larger dendrites (Fig. 2). Heavily labeled apical dendrites often could be traced into the molecular layer especially when viewed under darkfield illumination and occasionally they could be seen to divide into 2 or more divergent branches. Bundles of corticospinal axons and their local collaterals were also intensely labeled and could be traced through the underlying white matter, internal capsule, cerebral peduncle, medullary pyramid, and when present, across the pyramidal decussation to the site of spinal cord section and HRP application (Figs. 2, 3).

Laminar distribution of HRP-labeled neurons

In every one of the individual cases, HRP-labeled neurons were found in cortical layer V and only in layer V.

Within layer V, however, labeled somata often appeared to be arranged almost vertically; that is, with several labeled neurons one nearly above the other (Figs. 2, 4). This vertical "stacking" of the labeled neurons within layer V showed considerable variation across species. The greatest stacking was found in bushbaby where as many as 12 (an average of 9) vividly stained neurons could be seen one above the other within layer V. The least was seen in rhesus macaque where rarely more than 4 neurons appeared to be stacked (Fig. 4).

Unlabeled neurons in layer V

Even in the densest zone of labeled neurons, the zone in which the stacks of HRP-labeled neurons within layer V often appeared to fill the entire height of the layer, inspection of Nissl-counterstained sections revealed that a large number of entirely unlabeled pyramidal-shaped neurons were also present (Fig. 5). In each animal, the shape of the somata of these unlabeled neurons ranged from pyramidal to tear-drop and therefore, the form of the unlabeled somata did not distinguish them from the labeled somata. Although we did not measure their size, the non-labeled neurons appeared to be considerably smaller than the labeled neurons in every case.

Interhemispheric asymmetry among labeled neurons

The relative number of labeled neurons in the hemisphere ipsilateral to the spinal cord hemisection was not intended to be a quantifiable result in this report because the depth of the hemisection beyond the midline of the cord and the spread of HRP into the uncut side of the cord were not controllable variables. Nevertheless, 2 inescapable features of interhemispheric distribution were revealed by counting the numbers of labeled cortical neurons found in the

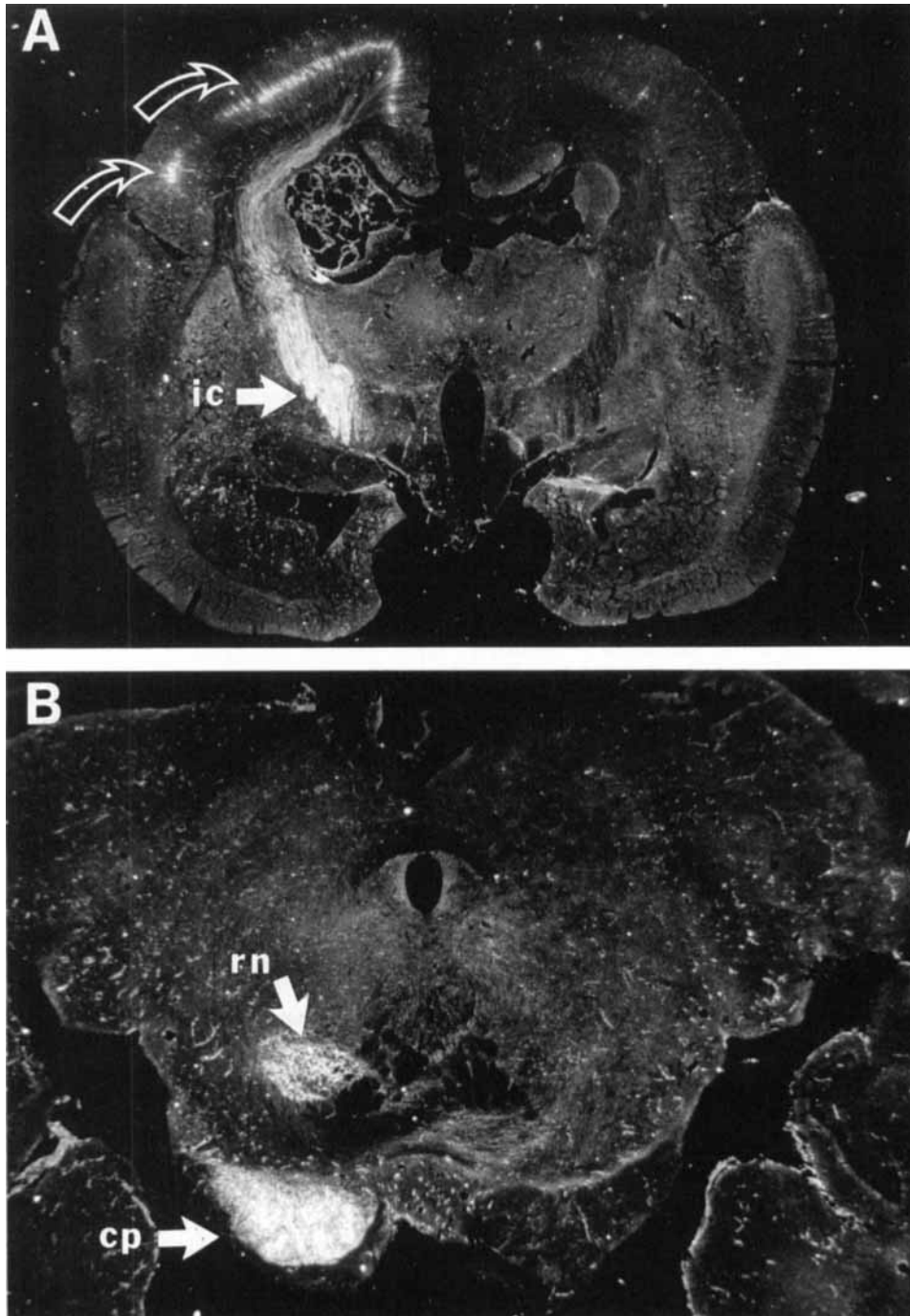


Fig. 1. Darkfield photograph through the forebrain $\times 6$ (A) and midbrain $\times 20$ (B) of marmoset monkey following application of HRP to a right spinal hemisection. Note the large number of HRP-labeled neurons in the contralateral (left) cerebral cortex in A and in the red nucleus (rn) in B and the intensity of fiber labeling in the internal capsule (ic in A) and cerebral peduncle (cp in B). The completeness of

retrograde filling of individual somata, axons, and dendrites both of cortical and noncortical neurons served as within-individual controls for the efficacy of the technique and lent assurance that no obvious species-specific variation was affecting HRP transport (also see Nudo and Masterton, '88). Hollow arrows in A indicate 2 separate regions of labeled cortical neurons.

hemispheres contralateral and ipsilateral to the spinal cord hemisection.

First, the overwhelming majority of labeled neurons were observed in the contralateral hemisphere in 18 of the 22 species. In these "typical" species, the proportion of HRP-

labeled neurons in the contralateral cortex exceeded 90% of the total in both hemispheres; that is, the ratio of contralateral to ipsilateral labeling was more than 9 to 1 (e.g., see Fig. 1). In the cortex of 2 of these cases (1 bushbaby, 1 rat) both with complete hemisections and no visible extracellular

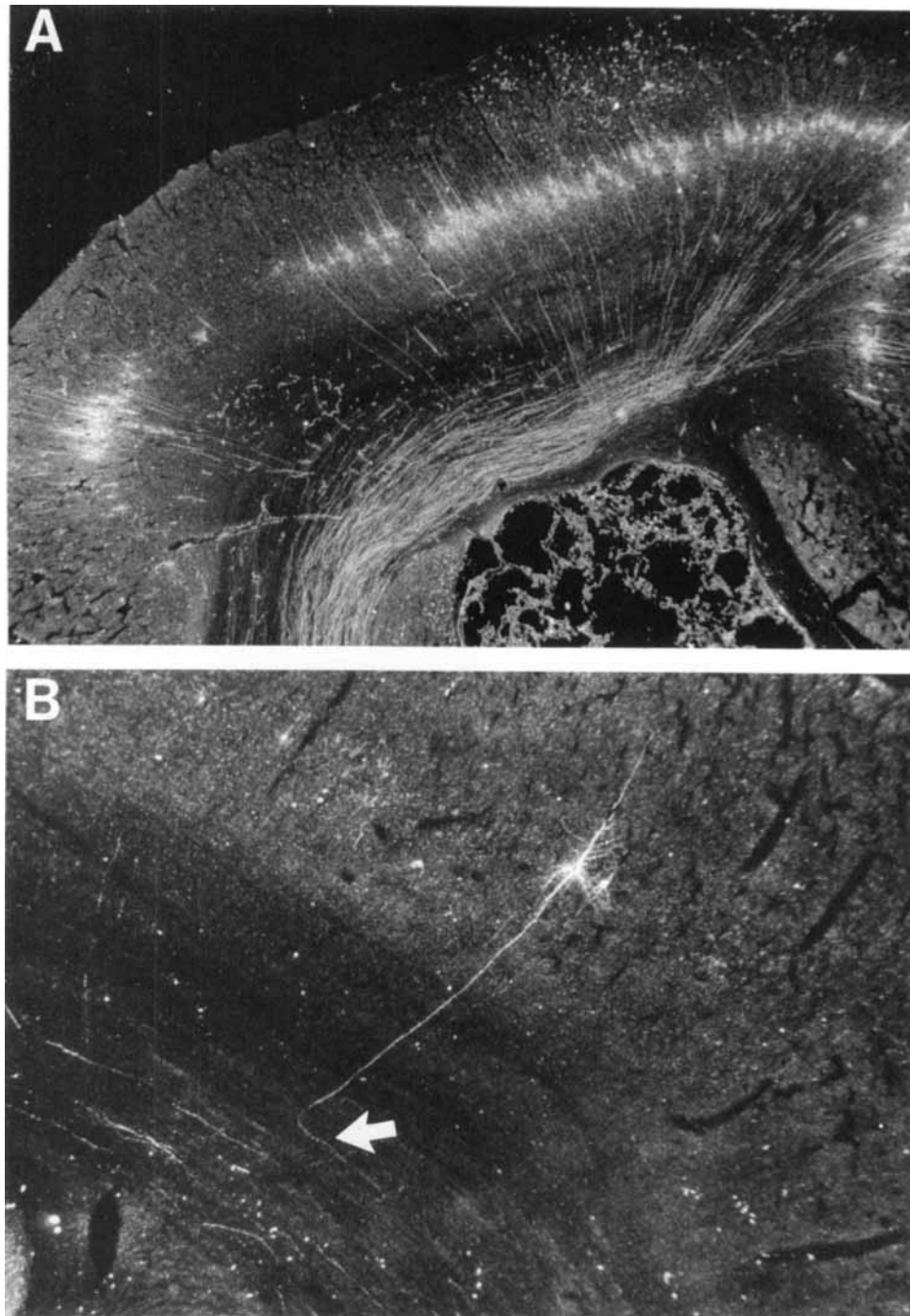


Fig. 2. **A:** Higher-power photograph of HRP-labeled neurons in the contralateral cortex of the same section shown in Figure 1A. $\times 20$. Note 2 separate regions of labeled neurons and axons entering the internal capsule. **B:** Still higher-power photograph of a single neuron in the ipsi-

lateral (right) cortex shown in Figure 1A. $\times 36$. Apical and basal dendrites and axons entering the internal capsule (arrow) were routinely labeled.

HRP either in the dorsal column nuclei or crossing the midline of the cord, the ratio of labeled contralateral neurons to total labeled neurons exceeded 99.5%; that is, a 200 to 1 contralateral to ipsilateral ratio.

Second, 4 of the 22 remaining species had the majority of labeled neurons in the cortex *ipsilateral* to the site of spinal cord application: the mole and each of the 3 species of hedgehog. In these 4 species, again with complete hemisec-

tions and HRP application closely confined to the sectioned side of the cord, the percentage of labeled neurons in the contralateral cortex averaged 33%, that is, a ratio of 2 *ipsilateral* neurons for 1 *contralateral* neuron (Fig. 6). These 4 exceptional species are each members of the order Insectivora.

Only 1 species of the total of 5 species of Insectivora was *not* exceptional—the least shrew (*C. parva*). In the least

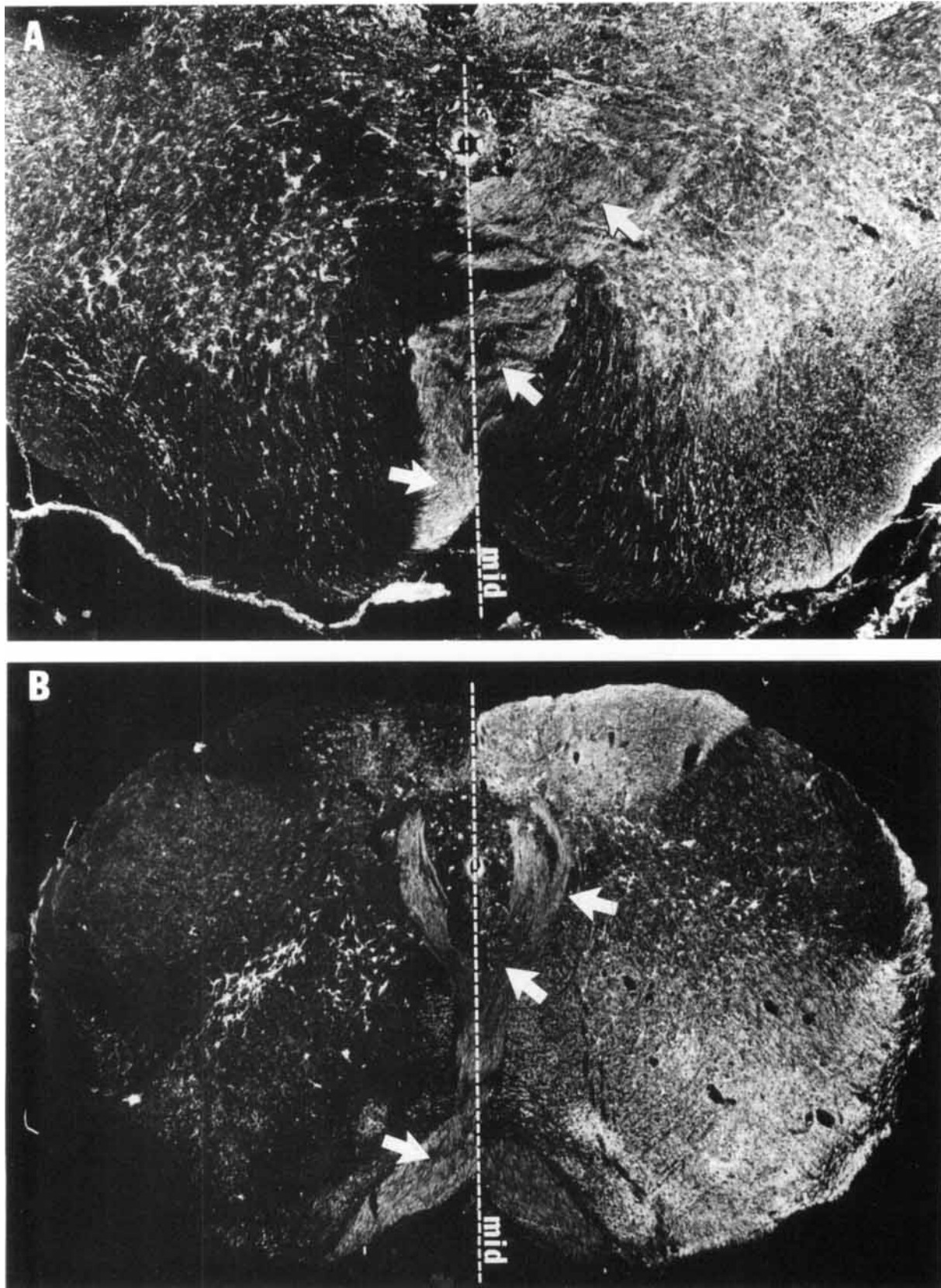


Fig. 3. Labeling of pyramidal decussation in a primate (A) and a rodent (B). $\times 30$. In A, labeled fiber bundles (arrows) cross midline toward the lateral funiculus whereas in B, fiber bundles (arrows) cross midline toward the dorsal funiculus.

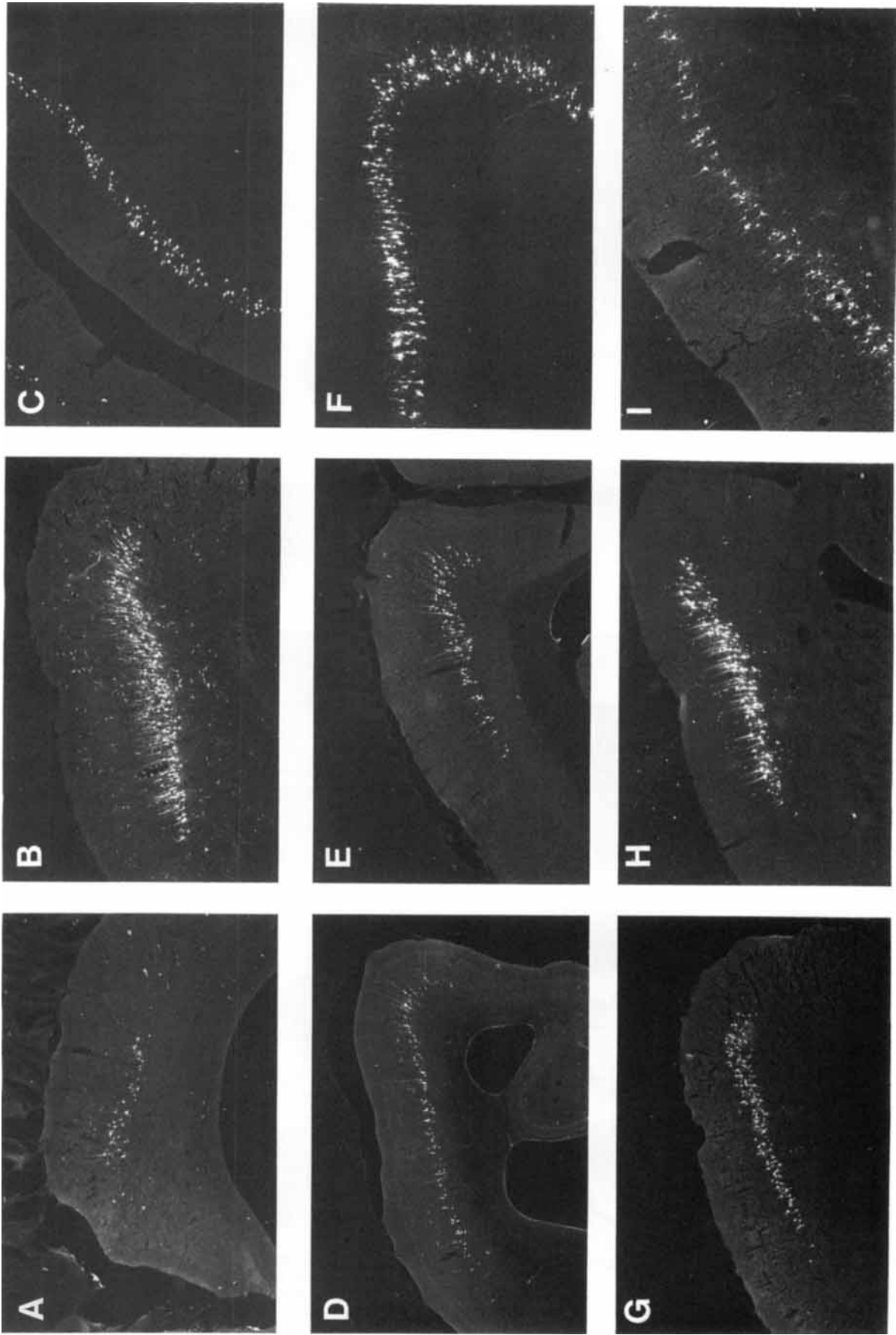


Fig. 4. Neocortical labeling in 9 genera of mammals. These low-power photos were made by projecting the actual tissue sections directly onto photographic paper. $\times 13.5$. The photos are arranged so that the side of the spinal cord hemisection is always on the right side. Therefore, each photo is the contralateral cortex except for the European hedgehog (see text). **A:** European hedgehog. **B:** Gray squirrel. **C:** Raccoon. **D:** North American opossum. **E:** Rat. **F:** Cat. **G:** Tree shrew. **H:** Ground squirrel. **I:** Rhesus macaque.

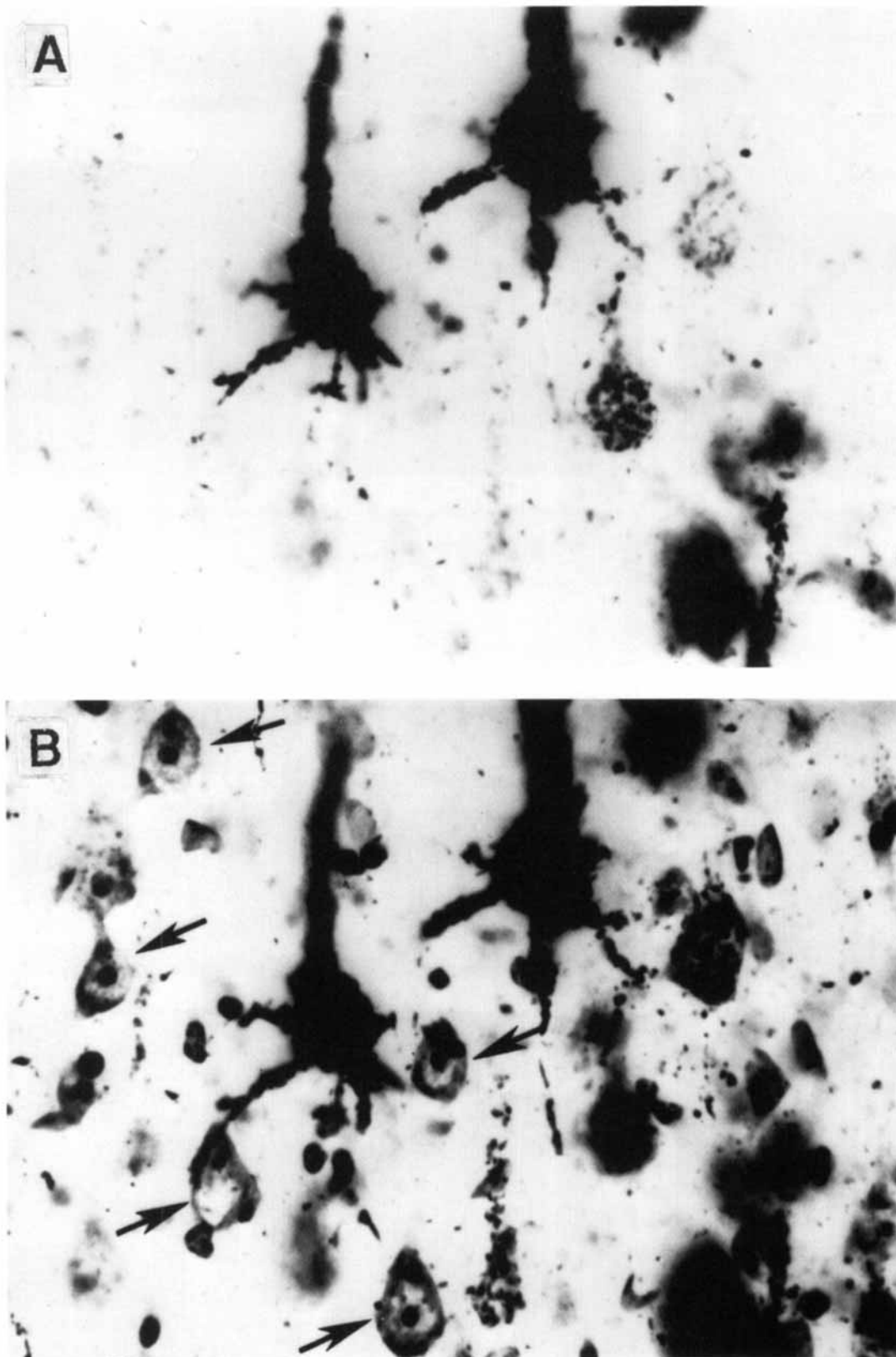


Fig. 5. Two high-power photomicrographs of the same set of layer V neurons (in gray squirrel) illustrating the relative concentration of labeled and unlabeled neurons. $\times 640$. **A:** Tissue photographed through a blue filter to suppress the cresyl-violet-stained neurons—only HRP-

labeled neurons appear. **B:** Same tissue photographed through a red filter in which both the labeled and unlabeled cresyl-violet-stained neurons appear black. Arrows in lower photo indicate some unlabeled neurons that do not appear in upper photo.

shrew, the proportion of contralateral neurons to total labeled neurons was over 90%, or 9 to 1. Therefore, the least shrew was indistinguishable from the typical or non-Insectivora ratio. Conversely, only 1 species that is *not* a member of the order Insectivora showed a larger number of ipsilateral than contralateral labeled neurons: the rock hyrax (*Procavia capensis*). Unfortunately, the single representative of this species did not fulfill the requirements of a complete hemisection without extracellular HRP in the dorsal column nuclei. Therefore, it is not included in this report. We mention it here only because the predominant ipsilaterality of its CST confirms Verhaart's ('67) observations using orthograde technique.

The distribution of the few labeled neurons that were found in the ipsilateral hemisphere of the 18 typical species was a greatly attenuated mirror-image of the distribution found in the contralateral cortex (e.g., see Fig. 7). That is, no ipsilateral neurons were found in areas not also labeled contralaterally, though many contralateral neurons were found in areas not labeled ipsilaterally. Similarly, the distribution of labeled neurons in the contralateral cortex of the atypical hedgehogs and mole was an attenuated mirror-image of the distribution found in their ipsilateral cortex (Fig. 6). However, once again, because it was not possible to tell if these lesser numbers of labeled neurons in the non-dominant hemisphere might not have been labeled by the tracer crossing the cord's midline, no conclusion regarding the exact proportion of projections from the non-dominant hemisphere can be made in any of the present cases. We return to this question below.

However, in the figures that follow, the distribution of labeled neurons is illustrated for the dominant hemisphere in each case: that is, ipsilateral to the spinal cord hemisection for the hedgehogs and mole and contralateral to the hemisection in all other species.

Distribution of HRP-labeled neurons across the cortical surface

Radial reconstructions of the lateral and dorsal views of the cortical surface in the dominant hemisphere of each of the species are shown in Figures 8 through 13. These figures illustrate a general uniformity among the species. We begin with these points of similarity before describing the variations.

There are at least 4 broad features of the cortical distribution of labeled cells that were identical in each of the 22 species: *First*, labeled cells were invariably observed in neocortex and among telencephalic cortical structures, only in neocortex. Specifically, no labeled cells were ever seen in the hippocampus or ventral to the rhinal fissure, in pyriform cortex or in olfactory structures. Furthermore, no labeled cells were seen in the cingulate gyrus, although large numbers of labeled neurons were often seen in the ventral banks of the cingulate sulcus.

Second, no labeled neurons were ever observed at the occipital pole, or in the striate or peristriate areas in any species in which these areas do not surround the occipital pole.

Third, in species containing an unarguable temporal lobe (i.e., the primates alone), no labeled neurons were ever observed within it. No labeled neurons were present ventral to the Sylvian (or quasi-Sylvian) fissure, although many labeled neurons were seen buried in its medial bank—that is, in insular cortex.

Fourth, with the exception of the rodents (see below), no labeled neurons were observed near the frontal pole.

Restating these results in other words, despite the clear labeling of thousands of cortical neurons in the individual cases, the labeled neurons were always confined entirely to the parietal and posterior frontal cortex. In every case in which cytoarchitectonic areas could be defined with certainty, Brodmann's frontal Areas 8, 9, 10, 11, and 12, temporal Areas 20, 21, and 22, and occipital Areas 17, 18, and 19 were completely devoid of labeled neurons.

Similarities in cortical regions containing labeled neurons

In every case the regions containing high concentrations of labeled neurons could be easily circumscribed. The total area of cortex containing labeled neurons ranged from 1 mm² in least shrew to 1,018 mm² in raccoon with an average of 197 ± 14 mm² for the 22 mammals. In terms of percentage of total neocortical surface area, the area originating CST fibers ranged from less than 5% in *Monodelphis* opossum and least shrew to over 25% in raccoon with an average of $13.7\% \pm 0.3\%$ for the 22 mammals.

These regions included several specific cytoarchitectonic areas: namely, frontal Areas 4 and 6, and parietal Areas 3, 1, 2, 5, and 7. Most if not all of the variation in this correspondence was found among the non-primates and therefore possibly due more to the indistinct cytoarchitectonic borders in these species than to the presence of outlying labeled neurons.

While the borders of labeled regions were typically discrete and relatively abrupt at the level of resolution used here, the surface-density plots showed wide variations in density among the genera (right-most drawing in Figs. 8–13). However, the variability among individuals of the same species was slight and for present purposes can be considered to be negligible (e.g., see density plots of two gray tree squirrels in Fig. 10).

Either from the set of surface maps (Figs. 8–13) or from the set of density plots accompanying them it can be seen that there were at least 2 separate concentrations of labeled neurons in each case. That is, in each species at least 2 large and relatively segregated cortical regions containing HRP-labeled neurons could be distinguished by eye alone. For convenience, we will refer to these separate concentrations of labeled neurons as "Regions" (as opposed to cytoarchitectonic "Areas" or physiologically defined "Fields") and distinguish among multiple regions as "A," "B," "C," etc.

In these terms, a large "Region A" and a smaller "Region B" were present in each species. In the members of 3 of the 8 orders (primates, rodents, and lagomorphs) a third region was also present. On the basis of its differing location (see below), we refer to it as Region C in the primates and as Region C' in the rodents and lagomorphs.

Region A. Region A is the cortical region originating most of the fibers in the corticospinal tract. It is the largest of the 2 regions common to all species and was invariably located on the dorsal surface usually about halfway along the rostrocaudal extent of the cortex. Its area averaged 164.7 mm² which is 10.4% of the average area of neocortex. Region A invariably contained the largest number and percentage of labeled neurons (averaging $19,838 \pm 4,899$ labeled cells or $87\% \pm 2\%$ of the hemispheric total of labeled cells) and the highest average density of labeled neurons (300 ± 51 neurons/mm²) of any of the labeled regions in cortex. Therefore,

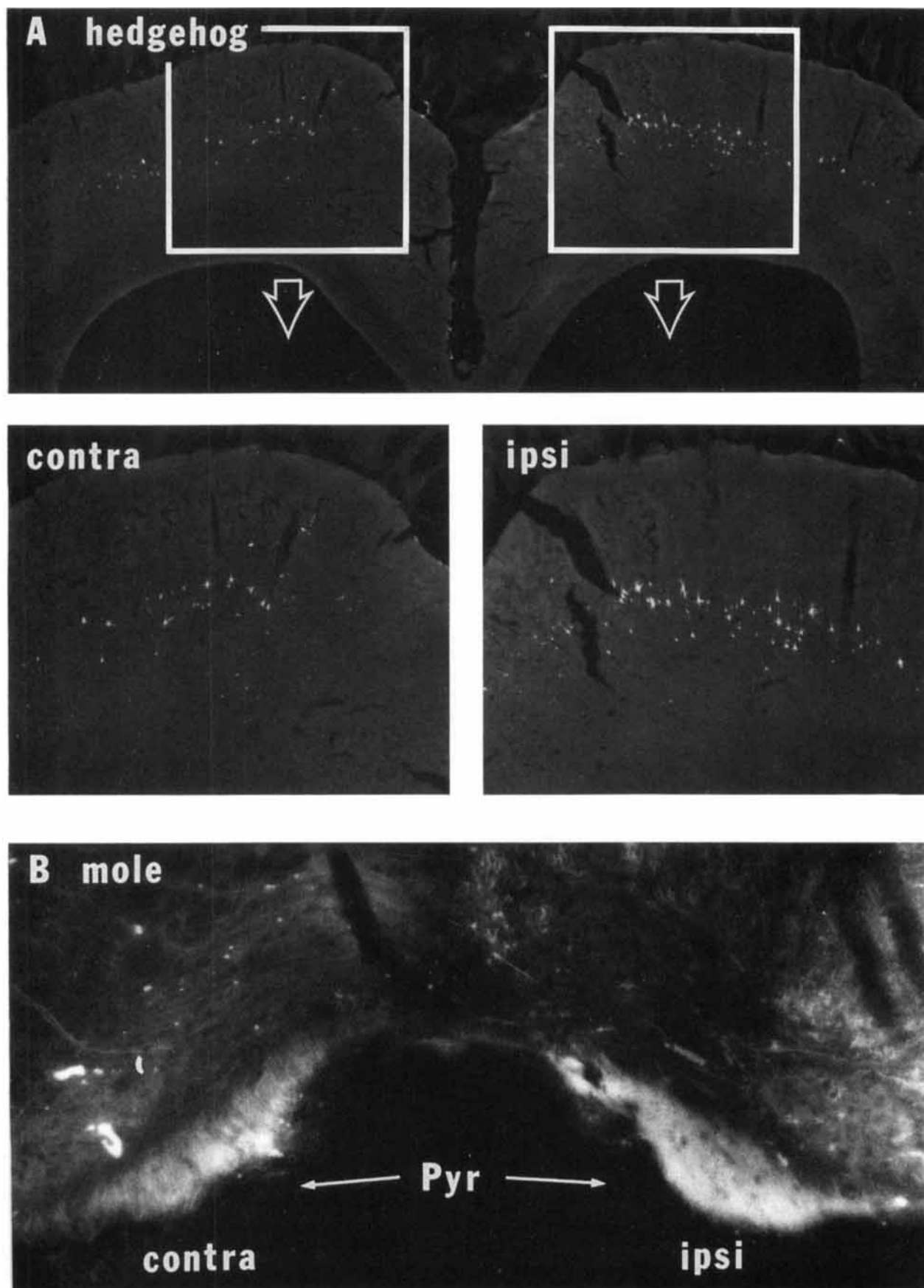


Figure 6

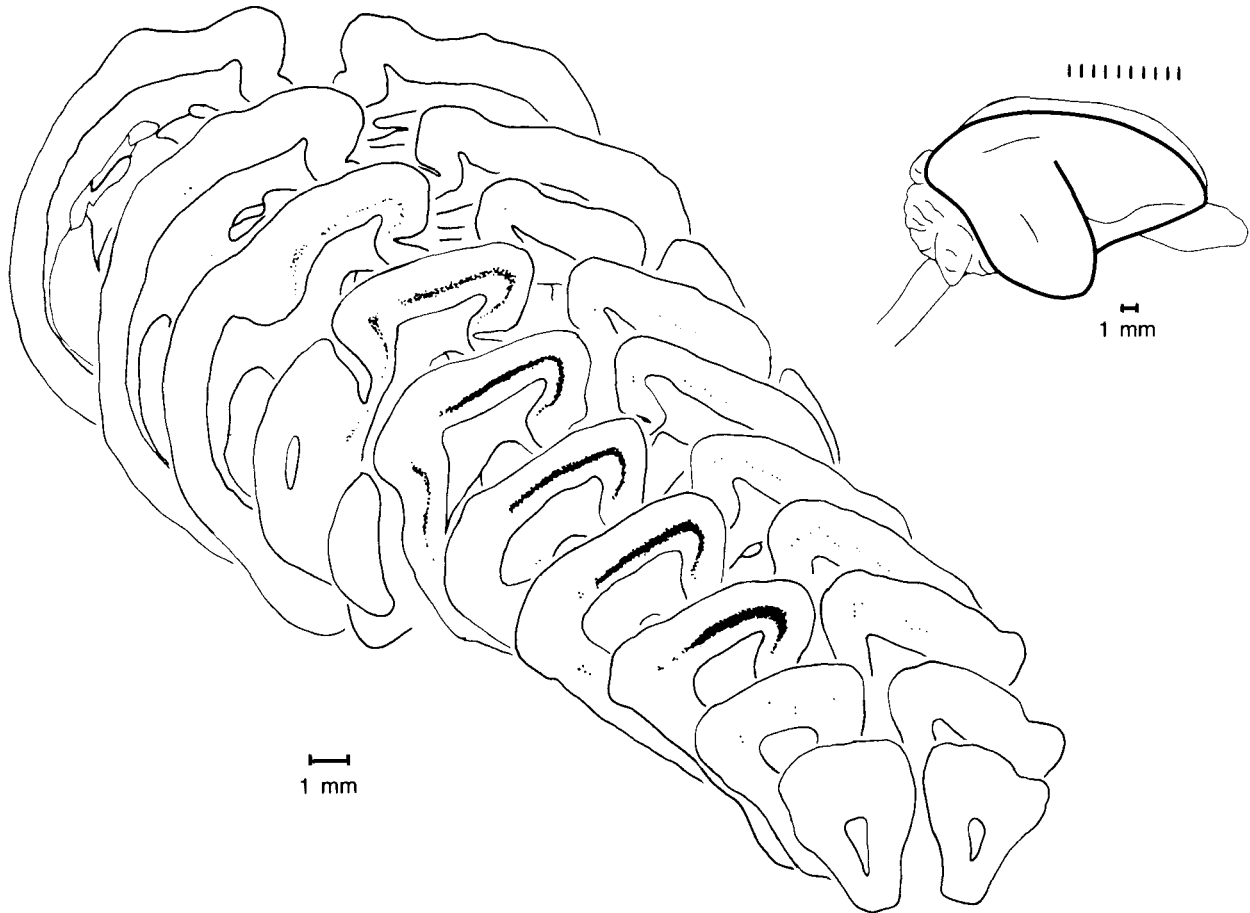


Fig. 7. Plots of labeled cells from a series of coronal sections through cerebral cortex of bushbaby. Each dot represents one labeled cell. Sections are 1 mm apart at levels indicated by the vertical lines in the lateral view of the brain.

Region A is the chief source of corticospinal axons in each species without exception.

Region A had the largest number of labeled cells in raccoon (89,006) and the smallest number in least shrew (320). It had the highest density of labeled cells in mole ($751/\text{mm}^2$) and the lowest density in *Macaca fascicularis* ($68/\text{mm}^2$). It was largest in absolute area in the raccoon ($912/\text{mm}^2$) and smallest in the least shrew ($0.9/\text{mm}^2$). Relative to total neocortical surface area it was largest in the raccoon (27.7%) and smallest in the *Monodelphis* opossum (2.5%).

Within Region A, labeled neurons were present in the banks and depths of any sulcus that was present (i.e., central sulcus and intraparietal sulcus in anthropoid primates; cruciate, coronal, and post-cruciate sulci in carnivores; orbital sulcus in *Didelphis*, etc.). With few exceptions (see below), Region A extended onto the medial wall and into the depths of the cingulate sulcus often as far as the medial

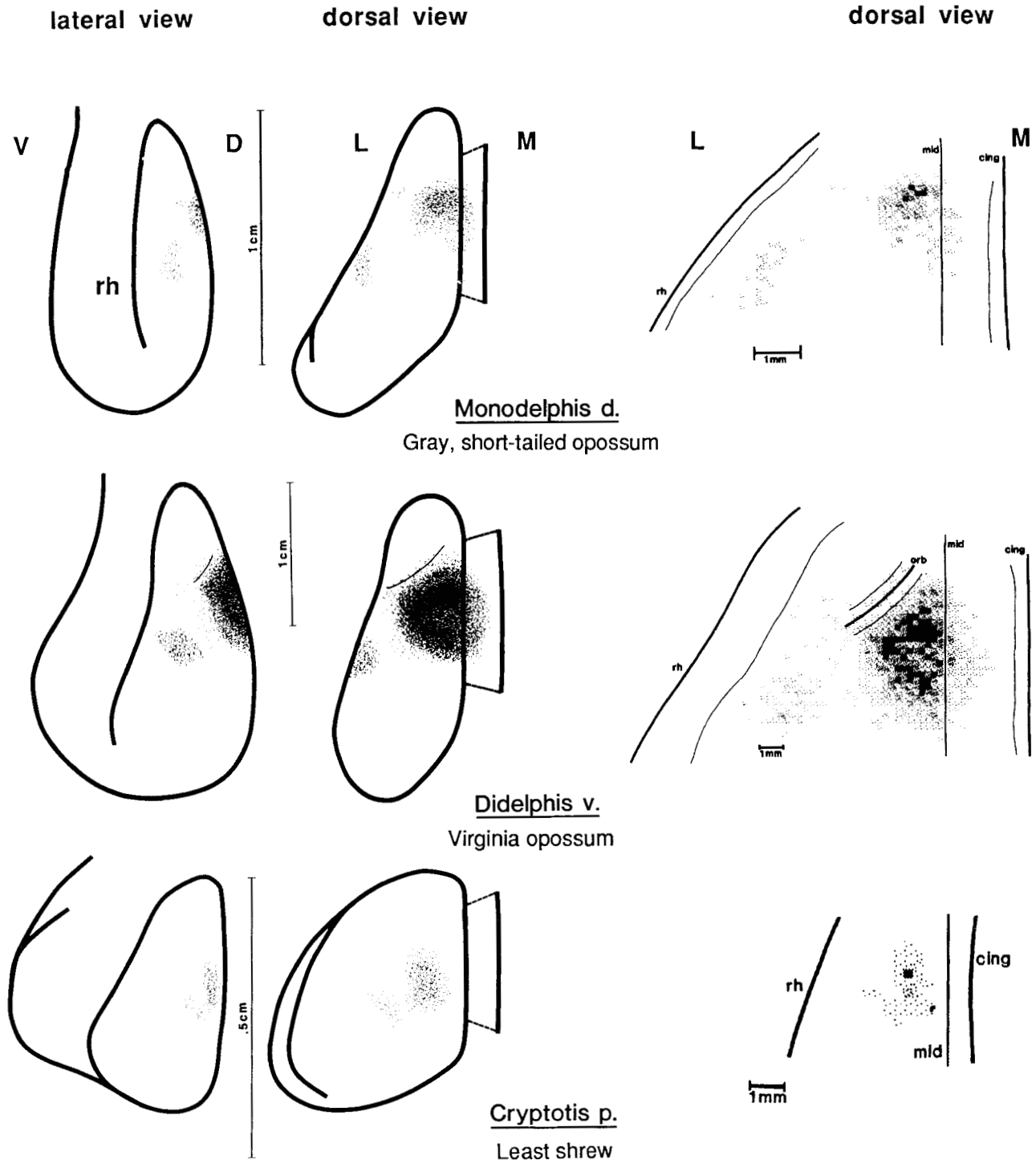
convexity of its ventral bank though it never extended onto the exposed gyral surface. In a lateral view, Region A usually extended ventrally about halfway to the rhinal fissure.

In the basal mammals (marsupials and insectivores), Region A corresponds approximately to the "parietal area" of Walsh and Ebner ('70), an area characterized by a lack of Betz cells and a relatively wide and granular layer IV in these animals. In the primates, where the cytoarchitectonic boundaries were clear, Region A includes Areas 6, 4, 3, 1, 2, and 5 and portions of 23 and 24. In carnivores, Region A contains at least Areas 6, 4, 3, 1, 2, and 5 (Hassler and Muhs-Clement, '64). (In the carnivore materials we were not able to delineate Areas 23 and 24 unequivocally.) Finally, in rodents, Region A is contained primarily within the "lateral agranular" area of Donoghue and Wise ('82) and extends laterally into the adjacent granular (SI) cortex.

Region B. The second region of high-density labeling common to all cases, Region B, was found more lateral and

Fig. 6. A: Low-power photomicrograph ($\times 16$ above, $\times 30$ below) through the left and right cerebral cortex of European hedgehog showing greater number of labeled neurons *ipsilateral* to right spinal cord hemisection. Although the overwhelming majority of labeled neurons were observed in the contralateral cortex in 18 of the 22 species,

4 exceptional species (the mole and each of the 3 species of hedgehog) had the majority of labeled neurons in the ipsilateral cortex. B: Medullary pyramids in mole following right spinal hemisection (Cobalt-DAB processing). $\times 80$. Note heavier fiber labeling in ipsilateral (right) pyramid.



Figs. 8-12. **Left:** Lateral and dorsal reconstructions of the cortex, rostral up, showing the location of HRP-labeled neurons contralateral to spinal hemisection for each of 22 genera. **Right:** Density map showing number of labeled cells per $80 \mu\text{m}^2$ of cortical surface is also shown for each case. In this map, sulci and fissures are exposed with depth of rhinal

fissure (rh) on far left and midsagittal convexity (mid) and crest and depth of cingulate sulcus on far right. Calibration mark is 1 mm for density map and either 10 mm or 5 mm for surface reconstructions. V: ventral; D: dorsal; L: lateral; M: medial; rh: rhinal fissure.

posterior to Region A and was invariably smaller in area. It had both a fewer number and a lower percentage of labeled cells (averaging $10\% \pm 1\%$ of the hemispheric total), a lower density of labeled cells (430 ± 71 cells per mm^2), and a smaller absolute ($2.5 \pm 8.6 \text{ mm}^2$) and relative area ($2.52 \pm$

0.4%) than Region A. It had the largest number of labeled cells in bushbaby (2,931) and smallest number in least shrew (69). It was most dense in mole ($356/\text{mm}^2$) and least dense in Rhesus macaque ($5.7/\text{mm}^2$). It had the largest absolute area in Rhesus monkey ($159/\text{mm}^2$) and the smallest in least

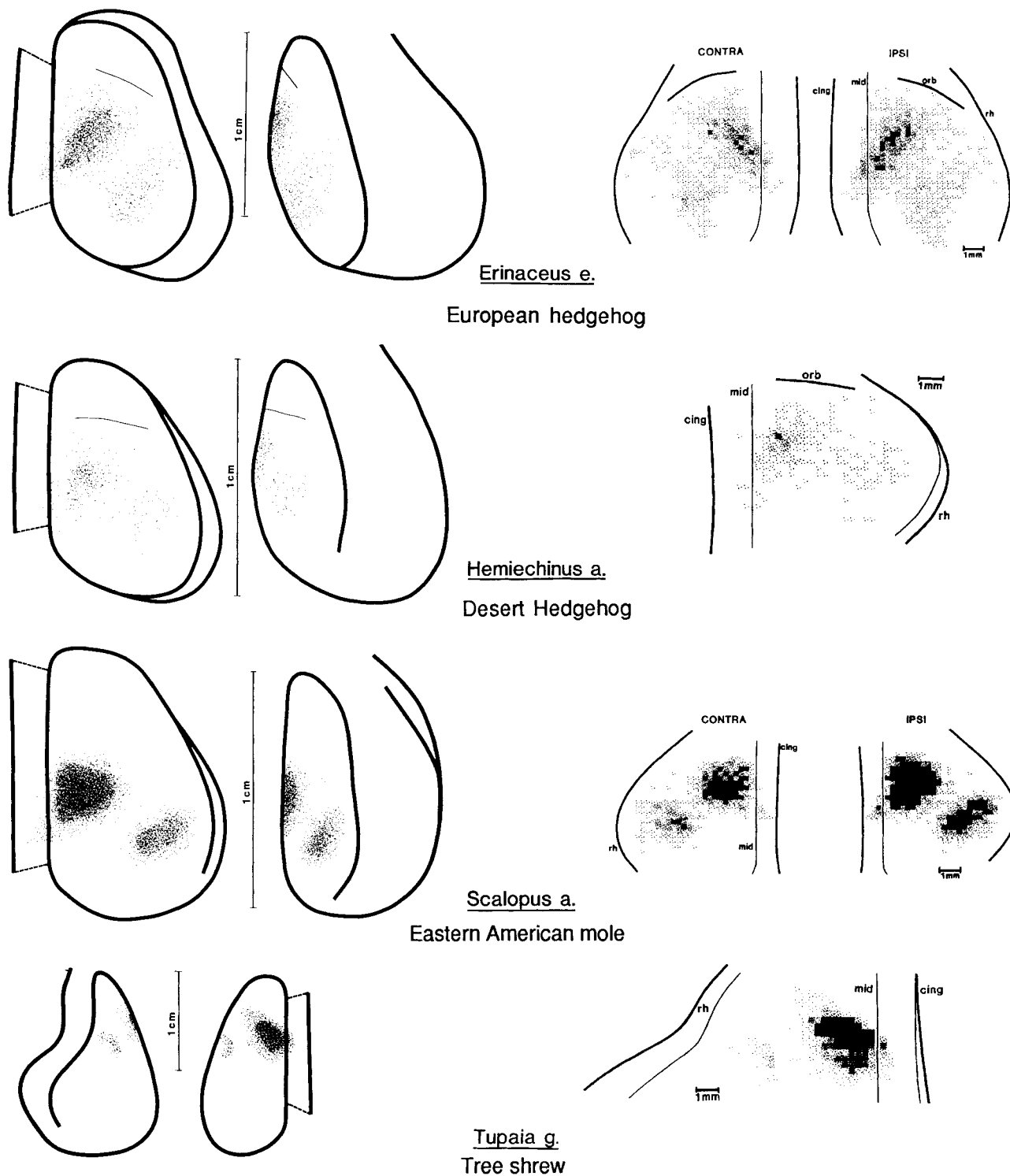


Figure 9

shrew and *Monodelphis opossum* ($0.4/\text{mm}^2$). It had the largest area relative to total neocortical surface area in mole (9.2%) and smallest in *Monodelphis opossum* (0.6%). In its area relative to the area of Region A, it averaged $26.7 \pm 3.6\%$ and this percentage was largest in mole (66.7%) and smallest in slow loris (9.2%). In each of these respects, Region B is subordinate to Region A in every animal in the sample.

Region B is located posterolateral to Region A, usually beyond but sometimes abutting the posterolateral boundary of Region A (see Figs. 8–13). It corresponds loosely with the second somatosensory field (i.e., SII) but is usually somewhat larger than electrophysiological maps of this field. In the less fissured brains (e.g., opossum, rat, hedgehog, etc.), the entire extent of Region B could be seen in either dorsal

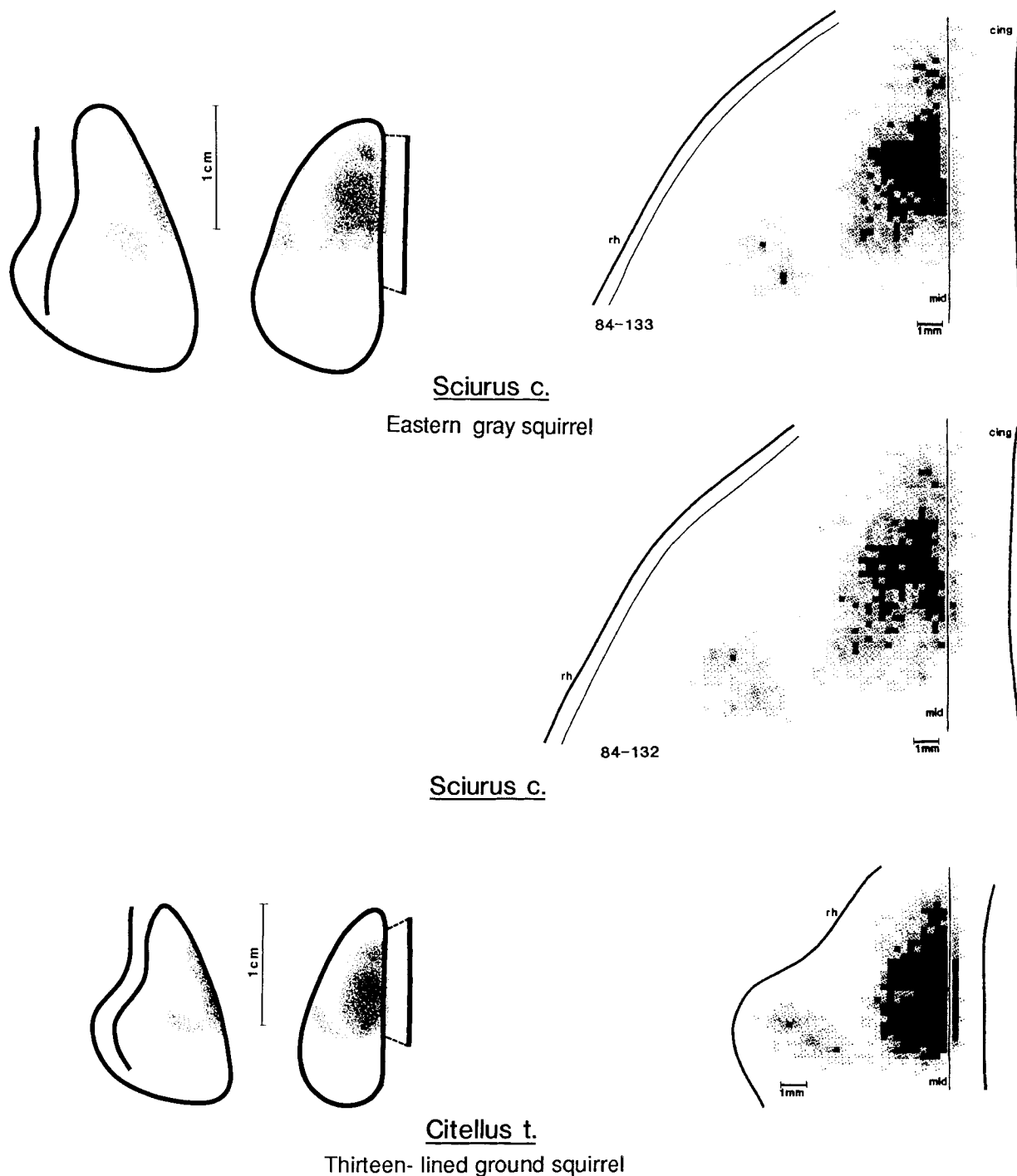


Figure 10

or lateral views of the cortical surface. In heavily fissured brains (e.g., anthropoids, raccoon, etc.), only part of Region B could be seen from the surface since the majority of the labeled cells were buried in the depths of the Sylvian (or pre-Sylvian or pseudo-Sylvian) fissure.

In the basal mammals (marsupials and insectivores), Region B appeared to be wholly contained within the insular area. In the primates, the largest part of Region B was found buried in the Sylvian fissure, in insular cortex. However, a

small part of Region B was located dorsal to the Sylvian fissure in each of the primates except for the squirrel monkey where Region B was completely hidden in the fissure. In the Old World monkeys, the exposed portion of Region B is contained within Area 7.

Region B was visibly segregated from Region A (that is, distinguishable as a "second" region by eye alone) because it is completely separated from Region A along more than 90% of its outer boundary in each animal except European

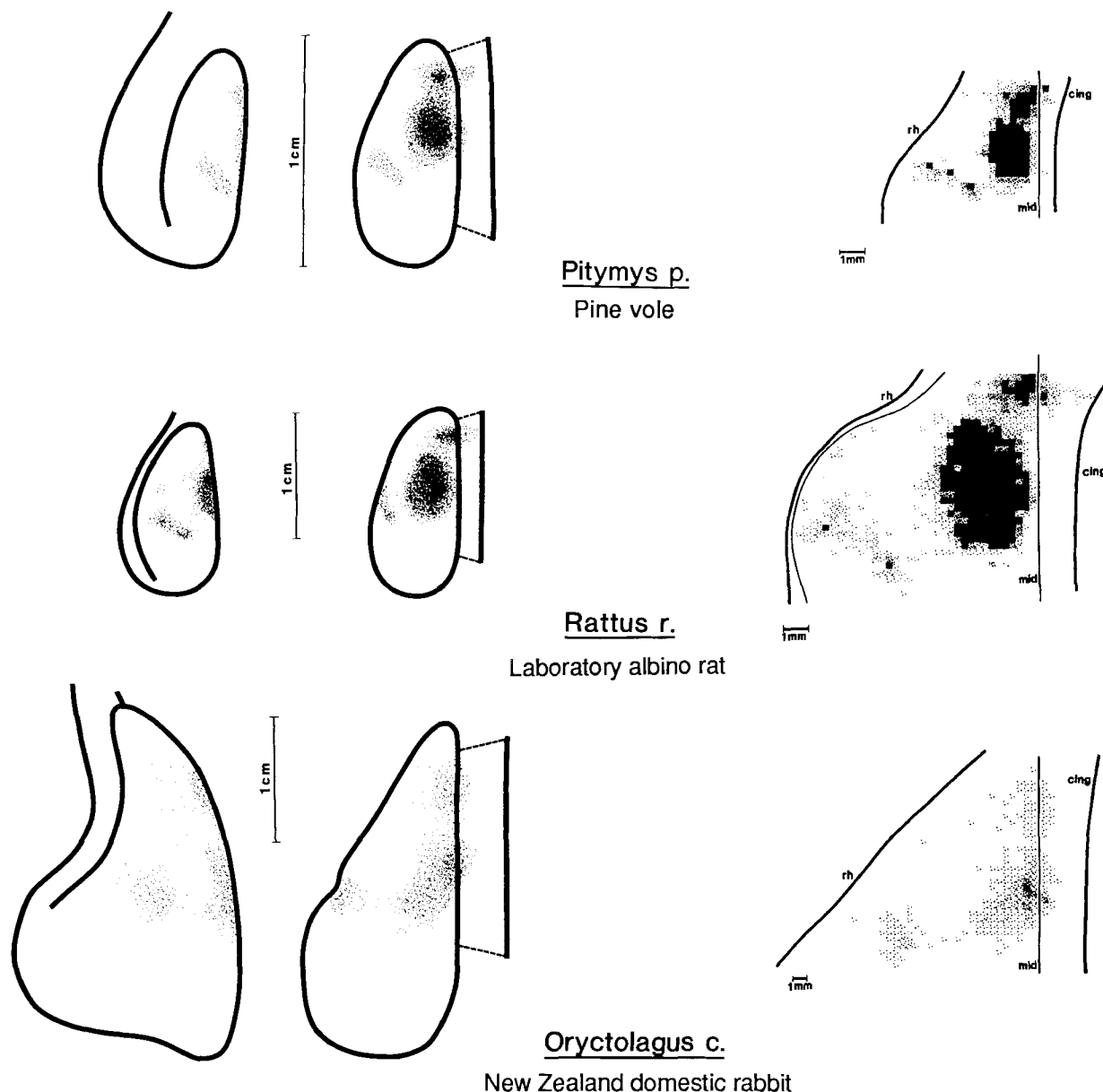


Figure 11

hedgehog. In this species, Region B's common boundary with Region A is separated along 80% of its total circumference.

Other regions of labeled cells. Because we wished to describe any other segregated regions of labeled-cell concentrations if they were present, we analyzed the density characteristics of Region A, Region B, and the zone between them in order to derive an unambiguous definition of their segregation. This analysis indicated that given the resolution of the density-maps, the visible segregation of Region B from Region A in each of the 22 species had 2 characteristics: 1) the *peak density* of Region B was always more than 24% of the peak density of Region A; and 2) the density of the cortical zone between Region A and Region B, the *valley density*, was always less than 50% of the peak density of Region B.

These "peak and valley" criteria accompanying the visible segregation of Region B from Region A provided the means

for determining if there might be still other "regions" present in some animals. As it turned out these criteria also served to distinguish a third region in some species.

Species with 3 discrete regions

In addition to the 2 chief regions (A and B) observed in every species, the "peak and valley" criteria indicated that a third "Region" was present in 12 of the 22 species. One (Region C) was present only in the 7 species of primates and the other (Region C') was found only in the 4 species of rodents and 1 species of lagomorph (rabbit). When the same quantitative criteria were used, no more than three regions were present in any species.

The peak-and-valley criteria indicate that both of these additional regions (C or C') are distinguishable from Regions A and B because they form a well-circumscribed zone with a relatively high-density of labeled cells and this zone is

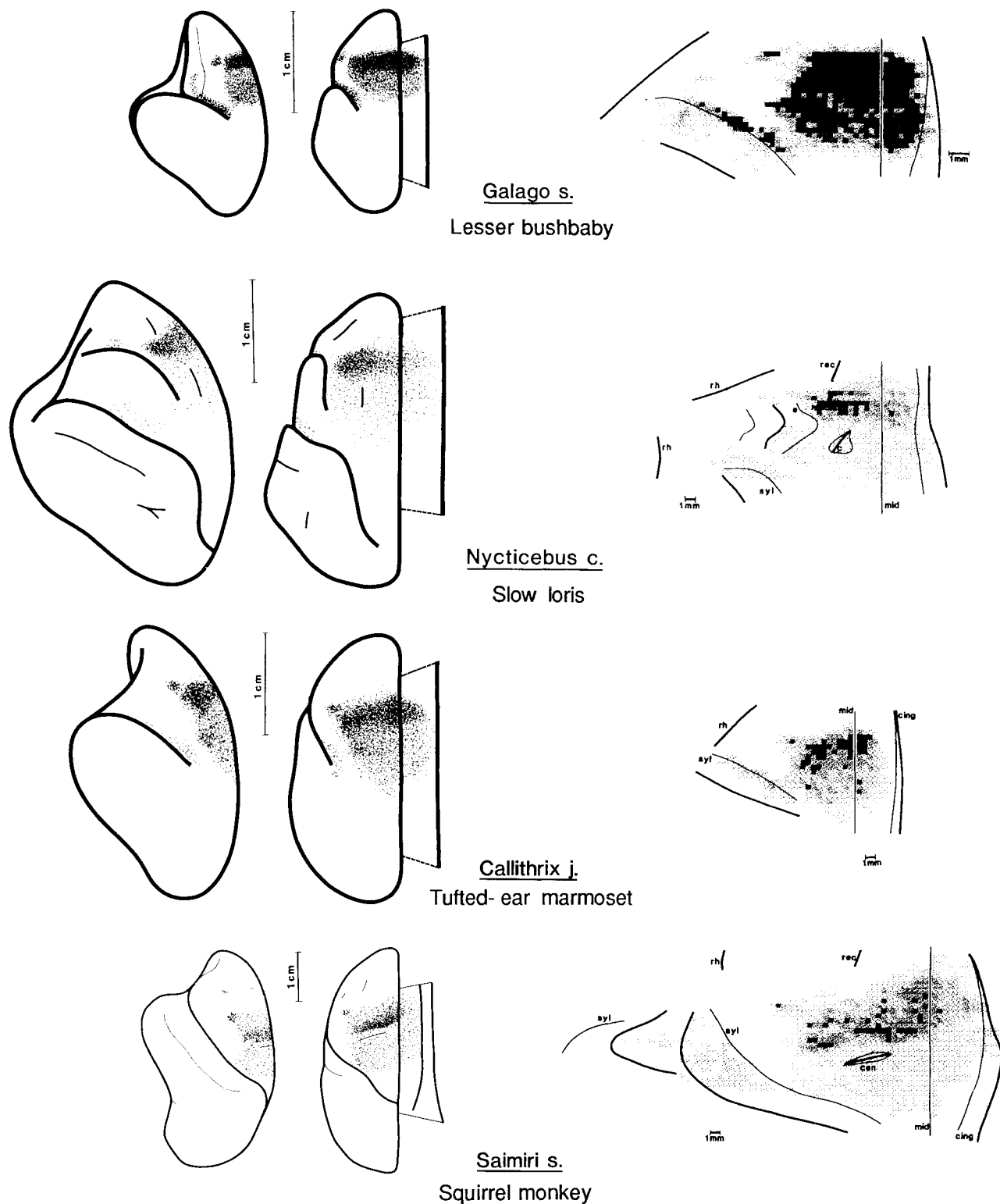
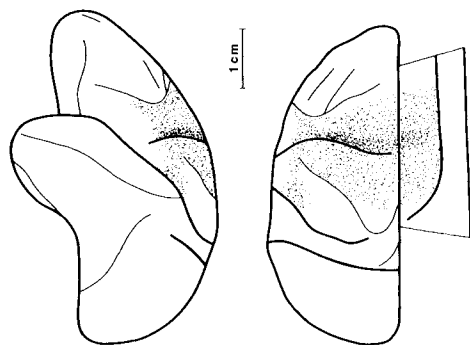


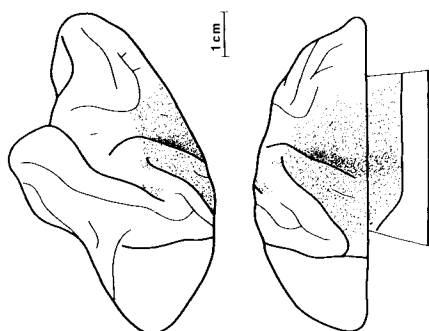
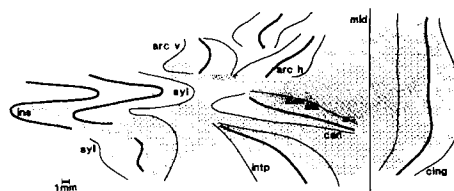
Figure 12

separated from both Region A and Region B by a band of low-density labeling (see Figs. 14–17, etc). Occasionally, these regions were entirely non-contiguous even at the relatively unrefined level of resolution used here: that is, they were separated by a band of cortex with no labeled cells whatever—far surpassing the “valley” criterion.

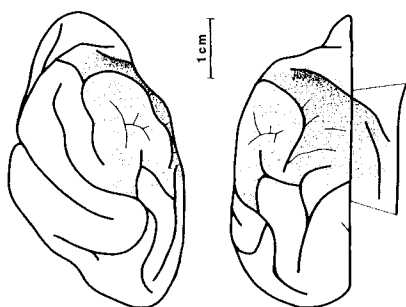
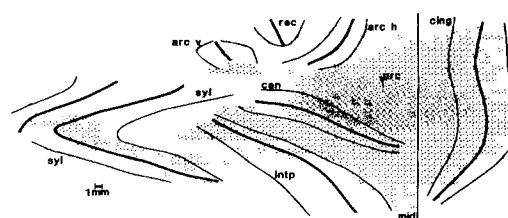
Region C. The first of these “additional” regions was a small one located lateral and rostral to Region A in the primates (see Figs. 15, 16). Among the 7 species of primates, it contained an average of 787 ± 144 labeled cells or about 2% of the hemispheric total with an average density of 166 ± 24 labeled cells per mm^2 . Region C had the most



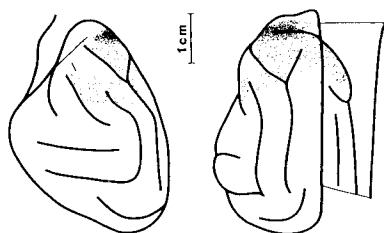
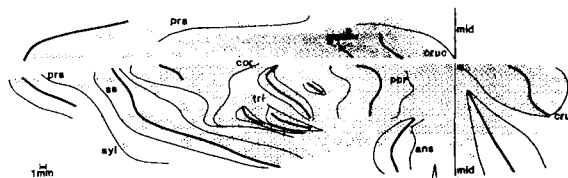
Cercopithecus a.
Green monkey



Macaca m.
Rhesus macaque



Procyon l.
Raccoon



Felis c.
Domestic cat

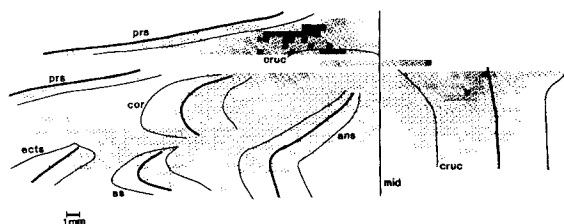


Fig. 13. Abbreviations same as Figures 8–12. Gaps in density plots for raccoon (*Procyon l.*) and cat (*Felis c.*) are artifacts of reconstruction, not gaps in labeled-cell distributions.

labeled cells in squirrel monkey (1,233) and least in rhesus macaque (754). It was most dense in bushbaby (1,504 labeled cells per mm²) and least dense in fascicularis macaque (128 per mm²). In these respects, Region C is subordinate to both Region A and Region B.

Region C is contained entirely within the lateral part of Brodmann's Area 6. For example, in prosimian primates (bushbaby, slow loris) and in Old World monkeys (green monkey, macaques), Region C can be seen to be almost completely separated from Region A (e.g., Figs. 15, 16). In

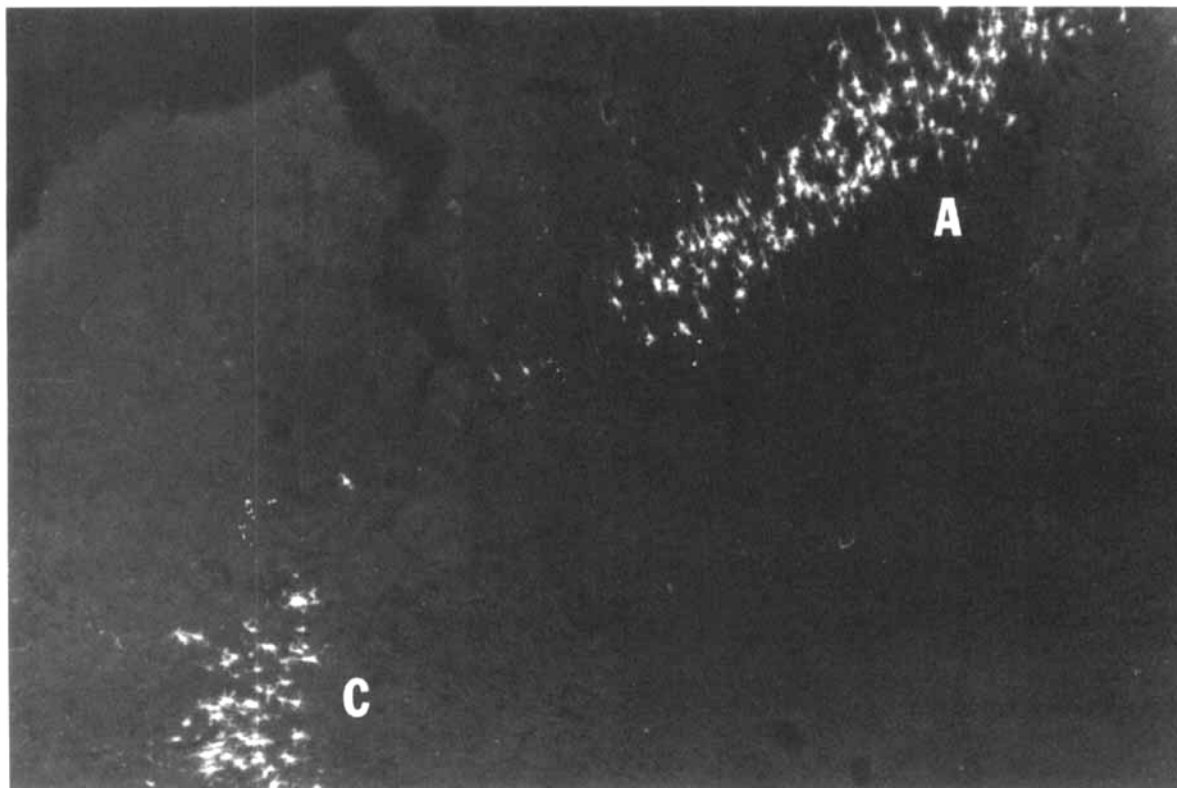


Fig. 14. Photomicrograph showing Regions A and C in a prosimian primate (bushbaby). $\times 42$. Region C was present in each of the 7 primates, but in no other species of mammals.

the New World monkeys (marmoset and squirrel monkey) Region C nearly abuts Region A though still satisfying the criteria for a separate Region both with a sharply higher density of labeled cells within it and its separation from Region A by a zone of low-density labeling (see Fig. 15).

Region C is visible in a lateral view in the prosimian bushbaby and slow loris (Figs. 15, 16) but it is partially or entirely hidden in the caudal bank of the inferior arcuate sulcus in Old World monkeys (Figs. 15, 16).

A few labeled neurons in the same approximate position as Region C were found in a few other animals but they were far too few and too scattered to meet either the "peak" or valley criterion of a "region." Therefore, based on the same level of resolution and the same criteria used to recognize Region B as separate from Region A, Region C can be discriminated in each of the 7 primate species but in none of the other 15 species of mammals.

Region C'. An apparently different third region, Region C', can be recognized in the frontal cortex of each of the 5 species of rodents and also in the one species of lagomorph (rabbit)—both orders of the superorder Glires (Simpson, '45; Heffner and Masterton, '80; see Figs. 10, 11, 17). Region C' averaged $1,169 \pm 193$ labeled cells with an average density of 276 ± 101 labeled cells per mm^2 . It is largest (1,873 labeled cells) and densest ($1,601 \text{ cells/mm}^2$) in rat and smallest (553 labeled cells) and least dense (170 cells/mm^2) in rabbit. In these respects, Region C' is subordinate to Regions A and B.

Cytoarchitectonically, Region C' is contained within the medial agranular field of Donoghue and Wise ('82). At the present level of resolution, this frontal Region C' is not visibly isolated from Region A in any of the species in which it is present. However, it has a high-density of labeled cells

and it is separated from Region A by an area of low-density labeling that satisfies the same quantitative criteria differentiating Region A from Region B. Region C' is most obvious in rat in which it forms a well circumscribed, high-density area extending medially onto the medial wall of the frontal cortex (Fig. 11). Region C' is somewhat less obvious in the squirrels and rabbit in which it forms a moderately high-density area and extends only a short distance onto the medial wall (see Figs. 10, 11). Therefore, as in the case of Region C for primates, the same set of criteria distinguishing Region B from Region A also distinguished Region C' in the rodents and rabbit but in no other species.

Labeled cells not contained within one of the regions. Relatively few labeled cells were found outside of the regions just described. A few labeled cells (less than 0.5%) were sometimes found scattered around the perimeter of the regions, but the numbers of these fringe neurons were small and they were not found in the same location even in different individuals of the same species. For example, in 8 species (Virginia opossum, bushbaby, squirrel monkey, green monkey, cat, raccoon, rat, mole) several labeled neurons were observed anterior or anterolateral to the lateral limits of Region B. Since the density of these scattered cells was far too low to meet the "peak and valley" criteria used to distinguish the other regions in the same individuals, at present we consider them to be a minor and possibly variable source of corticospinal axons. They will not be mentioned further.

Variation in density within regions

The density of labeled neurons within a given region was rarely homogeneous. Zones of somewhat higher or lower

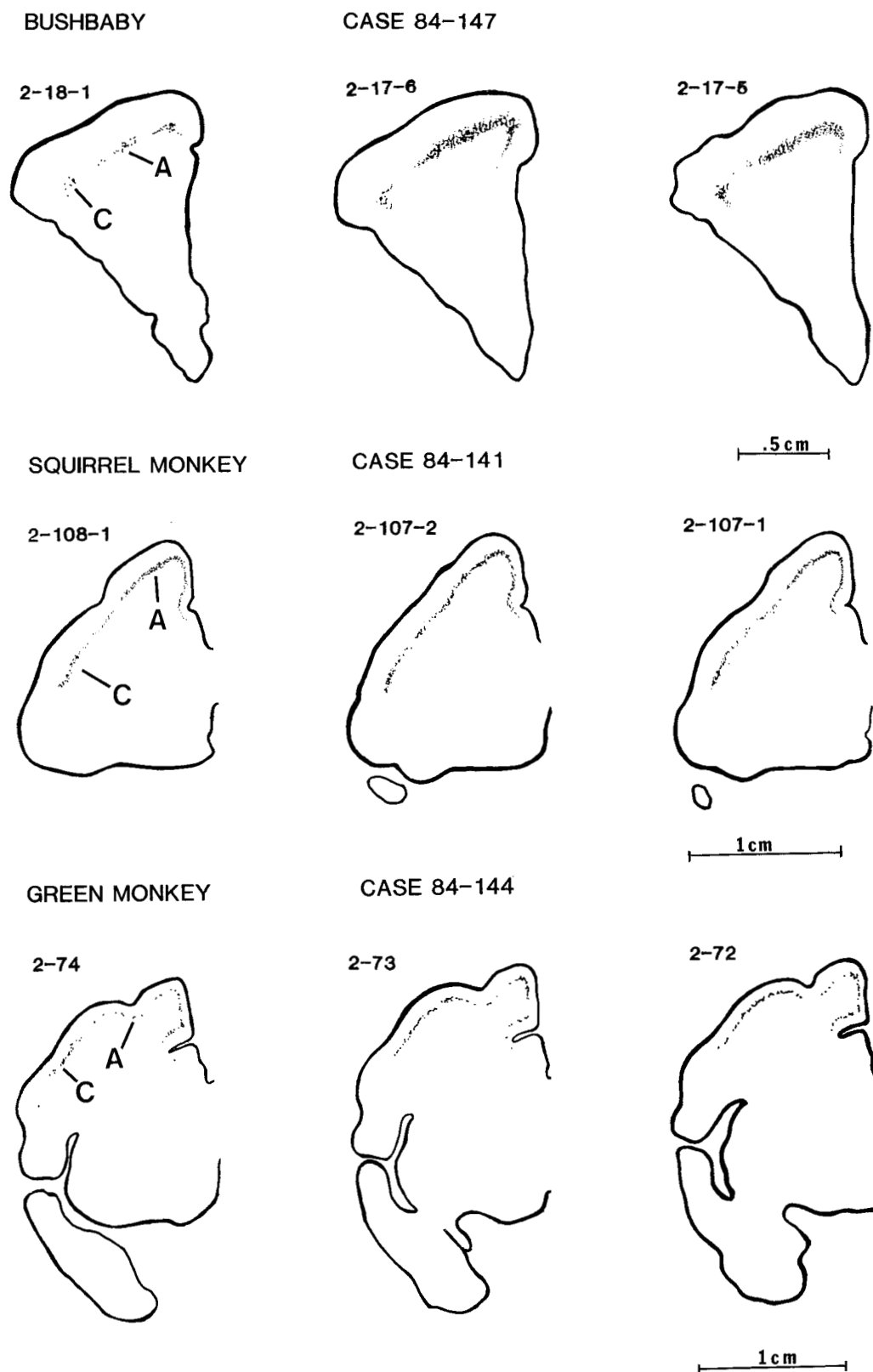


Fig. 15. Location of labeled neurons in 3 coronal sections through Regions A and C in three primates: prosimian primate (bushbaby), New World monkey (squirrel monkey), Old World monkey (green monkey).

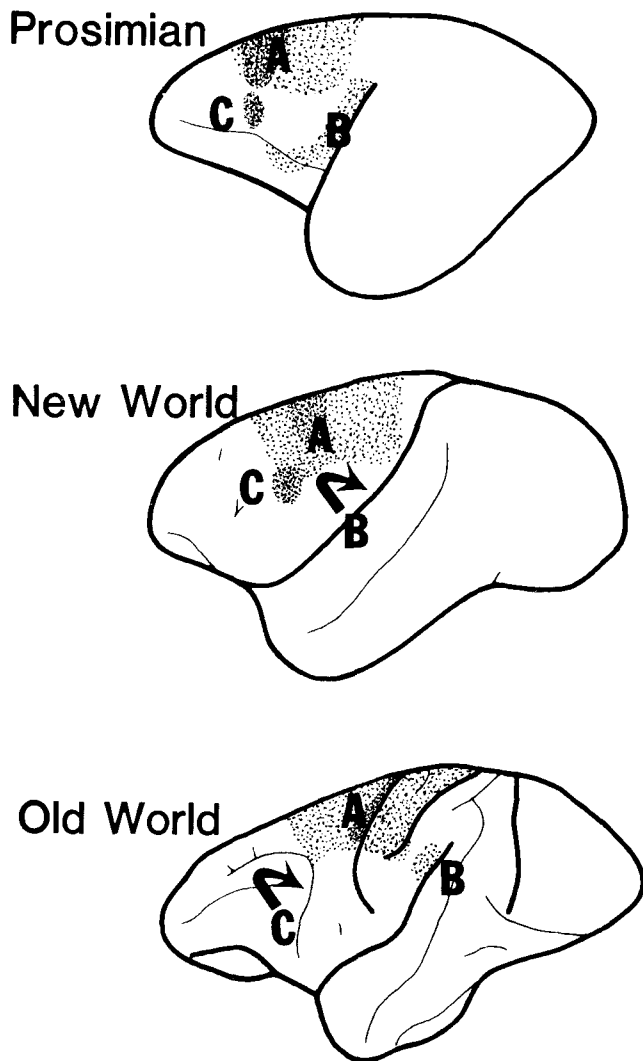


Fig. 16. Summary of the location of Region C in 3 families of primates. Curved arrow is intended to indicate that region is hidden from view in sulcus.

density were observed within the regions of most species and these non-homogeneities could be seen in the density maps regardless of the way in which the densities were displayed (i.e., either as absolute density, percentage of maximum density, normalized density, etc.). In general, these heterogeneities within regions are in the same locations in different individuals of the same species. However, these fluctuations in cell-density are much too small to meet the "peak and valley" criteria distinguishing between regions. For convenience, we will refer to these intra-regional variations as "zones."

High-density zones within Region A. There is considerable variation among the species in the location of relatively high- and low-density zones within Region A. However, related groups of species shared common zones. In *Marsupialia* (*Monodelphis* and *Didelphis* opossums), the density of labeled cells throughout Region A was relatively homogeneous (Fig. 8). However, a central core of somewhat higher density was usually observed surrounded by a fringe of lower density. Although this core was not well-circum-

scribed, the density in the rostral half of Region A was higher than in its caudal half.

In *Insectivora* (shrew, mole, 3 hedgehogs), the relatively low densities and relatively small brains of the species preclude a useful description of heterogeneity within Region A (Fig. 9). However, it was clear that in European hedgehog, a central core of higher density was present and it was oriented anterolateral to posteromedial, paralleling the orientation of the entire region. In the mole, however, the high-density zone extended throughout most of Region A dropping abruptly only near its borders (Fig. 9).

In *Scandentia* (tree shrew), the distribution of labeled neurons within Region A was similar to that of European hedgehog with a central core of high density oriented anterolateral to posteromedial, paralleling the orientation of the entire region (Fig. 9).

In *Primates* (bushbaby, slow loris, marmoset, squirrel monkey, green monkey, 2 macaques), a relatively small, well-circumscribed zone of high density was present within Region A, oriented approximately lateral to medial in prosimians (Fig. 12), somewhat posterolateral to anteromedial in New World monkeys, and rostral to the central sulcus (Fig. 12) and somewhat anterolateral to posteromedial in Old World monkeys in the rostral bank of the central sulcus (Fig. 13). The orientation of the high-density zone in primates paralleled the orientation of the central sulcus. This zone of high density was invariably contained within Area 4 and reached its highest density caudally, along its border with Area 3a. The density in this high-density zone decreased considerably as it extended onto the medial wall (with the exception of marmoset in which the peak density in the dorsal-most part of the medial wall was comparable to the density on the dorsal surface; see Fig. 13).

In each of the primates, Region A also contained a zone of low-density caudal to the high-density zone (Figs. 12, 13). This caudal zone lies within Areas 3, 1, 2, and 5 and within this zone, the density of labeling was quite homogeneous—within 1 standard deviation of its mean density.

Rostral to the high-density zone in primates, Region A contained a second, lower-density zone. This rostral zone is of substantial size in Old World monkeys and lies within Area 6. Labeled neurons in this zone were located on both the dorsal and medial surfaces. Further, there appeared to be a large variation in the relative size of this rostral lower-density zone among the primates. In the *Prosimians*, very few labeled neurons were located rostral to the Area 4/Area 6 border (see Fig. 12). In the *Old World monkeys*, a large zone containing many labeled neurons was found rostral to the Area 6/Area 4 border, especially close to and extending onto the medial wall. This zone appeared to be intermediate in size in the *New World monkeys*.

Finally, in the *Old World monkeys* (green monkey, 2 macaques), a narrow zone of very low density just caudal to the high-density zone could be seen in the density plots (Fig. 13). Upon close examination of sagittally sectioned and counterstained tissue, this label-sparse zone was found to be located just caudal to Area 4 in the trough of the central sulcus, apparently corresponding to Area 3a. While a similar low-density zone with approximately the same relationship to the high-density zone could be seen in each of the primates (Figs. 12, 13), this label-sparse zone was most apparent in the Old World monkeys.

In *Rodentia* (rat, vole, 2 squirrels), the distribution of labeled neurons within Region A differed between the muroid rodents (rat and vole) and the sciuroid rodents (tree

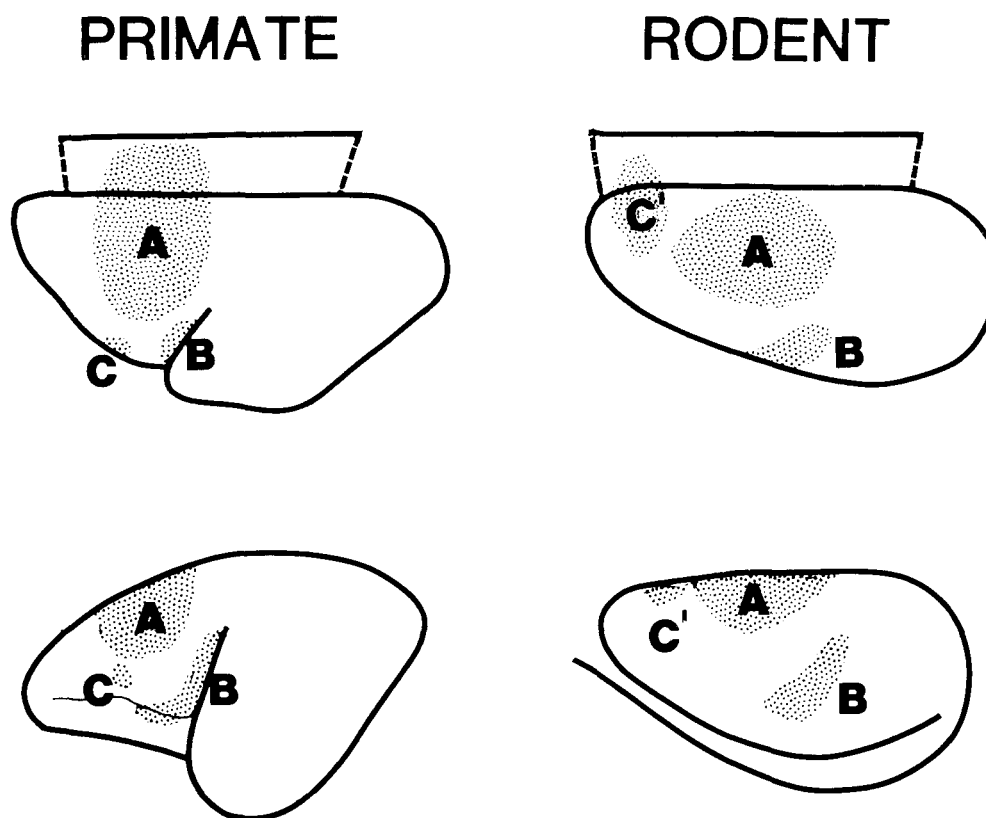


Fig. 17. Location of the 3 major regions of labeled neurons in cortex of primates (left) and rodents (right). Although Regions A and B are present in the same relative location in each of the species reported here, difference in location of Regions C and C' indicates that they are probably not homologous.

squirrel and ground squirrel). In the muroid rodents, a high-density elongated zone was present in the rostromedial portion of Region A, oriented rostromedial to caudomedial, similar to the orientation in European hedgehog and tree shrew (Fig. 10). In the sciuroid rodents (squirrels), the high-density zone appeared to be wedge-shaped, with its apex oriented caudomedially, and in addition, a second high-density zone was often present caudolateral to the wedge-shaped zone (Fig. 11).

Finally, in *Carnivora* (cat, raccoon), a relatively small and well-circumscribed high-density zone was found in the rostral-most part of Region A (Fig. 13). In cat, this high-density zone was located immediately rostral to the cruciate sulcus while in raccoon, it was located in the caudal bank of the sulcus.

Zones within Region B. In *Marsupialia*, Region B was completely isolated from Region A and contained a relatively homogenous distribution of labeled neurons (Fig. 8).

In *Insectivora*, Region B adjoined Region A along part of its circumference and contained a homogeneous distribution of labeled neurons. It was unusually large in insectivores, especially in European hedgehog where its areal extent (in the ipsilateral hemisphere) was nearly as large as Region A (Fig. 9).

In *Carnivora*, Region B formed an elongated zone of labeled neurons whose density was relatively homogeneous throughout (Fig. 13).

In *Primates*, Region B formed an elongated zone of labeled neurons, oriented caudomedial to rostromedial, its

largest part buried in the Sylvian fissure, i.e., in the insular cortex. Often, 2 zones of high-density were found within Region B, 1 located dorsomedially and the other located more ventrolaterally (e.g., see bushbaby, Fig. 12).

In *Rodentia*, Region B formed a somewhat elongated zone of labeled neurons, oriented caudomedial to rostromedial, segregated from Region A throughout most or all of its circumference (Figs. 10, 11). At least 2 and sometimes 3 distinct zones of high density were clearly evident within Region B in each of the rodent species. One was located near the dorsomedial limits of the region; another was located near its ventrolateral limits and sometimes a third was located between these 2.

Region C. Region C, distinguishable only in the primates, contained a single zone of dense labeling in each of the species studied (Figs. 12, 13).

Region C'. Region C' was distinguishable only in the rodents and in rabbit. In the rodents, it contained a single zone of high density near its center (Figs. 10, 11). In rabbit, the distribution of labeled neurons was relatively homogeneous throughout the region (Fig. 11).

DISCUSSION

The limitations on the present materials, the likelihood of significant overlabeling or underlabeling, and the restrictions on the sample of animals for inferences about other animals have been discussed in previous papers (Nudo and Masterton, '88, '89). Briefly summarizing these discussions

with specific reference to corticospinal neurons, 3 points warrant specific comment.

First, the present materials suggest the presence of more corticospinal neurons than have ever been reported before. Nevertheless, we feel that there is little chance that the materials were contaminated by HRP transport via blood or CSF or by extracellular spread of the HRP into the medulla. However, among those animals in which the CST terminates mostly in the highest cervical levels of the cord and with a significant number of corticobulbar axons terminating in the caudal medulla, the variation in the spinal cord hemisection between the C1 upper limit and the C2 lower limit permitted by our technique, together with the uncontrolled and possibly undetected spread of extracellular HRP rostralward, admits the possibility of imprecision despite the apparent invariance of the technique.

Second, due to the vast numbers of axons constituting the tract and the long distance required of the axonal transport, the possibility of underlabeling the corticospinal tract seems to us more likely than the possibility of overlabeling. Although a survival period of 3 days was chosen because that time period maximized the number of retrogradely labeled cortical cells in rats, this survival period was not increased for significantly larger brains nor decreased for significantly smaller brains. Therefore, an uncontrolled source of error may be present in the CST materials that is not present in the data obtained for shorter descending tracts (e.g., Nudo and Masterton, '89).

Third, it is important to note once more that the sample of mammals used here does not purport to be a random sample of mammalian individuals, species, genera, or families, though it is random for orders and still higher level taxa. Therefore, the variation within or among species, genera, or families is *not* usable for generalization or inference despite the strong temptation to use it as such. For these reasons, the following discussion is confined to conclusions which can be drawn at the ordinal level and above, with the one reference to family-specific variation, Soricidae (shrews) vs. other Insectivora (Erinaceidae, hedgehogs, and Talpidae, moles), made only because of its theoretical interest.

Number of discrete regions of cortex originating CST axons

In the mammals represented here, there appear to be either 2 or 3 separate regions of neocortex originating corticospinal fibers, not less and not more. Before considering the significance of this result, it should be noted that the number of separate regions certainly depends both on the resolution of the mapping and on the definition of "separate." However, a large number of other possibly relevant details are not relevant at all. For example, the absence of labeling of the face and head representations in the present materials does not mitigate the "separateness" of the regions found here. Nor does the obvious arbitrariness of the quantitative criteria used here to determine segregation. Although the number of regions clearly depends on the spatial resolution of the density maps and any criteria based on the density maps, the criteria used here to determine the presence of separable regions are unusually robust. For example, changing both criteria by 50%, that is, relaxing the "peak" criterion from 24% to 12% or raising the "valley" criterion from 50% to 75%, results in no increase (nor decrease) in the number of regions.

With the present sample including only 7 of the 20 ± 2 orders of mammals, it is perhaps surprising that the data

suggest a rather strong (though strictly statistical) conclusion. The uniformity in the number of regions (i.e., either 2 or 3 and neither more nor less) allows the strictly statistical inference that by using the same quantitative criteria and the same level of resolution adopted here, it is more likely than not (20 to 1) that *no* order of mammals exists which has either less than 2 regions or more than 3 regions originating CST fibers.

Statistical considerations aside, however, it can be noted that the present sample includes no orders with forepaws highly modified from the primitive pentadactyl condition—conspicuously absent are hoofed animals (Artiodactyla), whales (Cetacea), and bats (Chiroptera). Accepting the strong physiological (Lawrence and Kuypers, '68) and comparative (Heffner and Masterton, '75, '83) evidence that suggests that one unique contribution of the CST is related to independent movement of the digits, it would seem likely that if there are mammals with less than 2 regions of neocortex originating corticospinal fibers, they will most likely be found within the orders with highly modified or fused digits.

Presuming that the CST is indeed a key factor in digital dexterity, the possibility of a mammal having more than 3 regions might seem, at present, to be somewhat less likely if only because the most dexterous orders of mammals are already included in the sample (Heffner and Masterton, '75, '83). However, because an extra region originating corticospinal fibers is neither necessary nor sufficient for high digital dexterity (e.g., the dexterous raccoon has only 2 regions, the non-dexterous marmoset has 3), there would seem to be no reason excepting the statistical one for supposing that a mammal with 4 or more regions may not exist now or may not have existed sometime in the past.

Nevertheless, the present data suggest that the 2 or 3 regions of cortex originating corticospinal fibers, though often large, are invariably restricted, compact, and well-bounded; and within and among the orders represented, are stable both in absolute and relative location on the cortical surface.

Accepting the presence of only relatively few and well-bounded regions originating corticospinal axons, however, does not mean that there are an equally small number of physiological fields or somatic representations. Region A alone almost certainly contains at least 2, probably 3, and possibly 4 distinct physiological fields in each of the placental mammals—that is, at least a separate primary "motor" and primary "sensory" field, probably a "premotor" field, and possibly a "supplementary" field each with its own complement of CST fibers (Murray and Coulter, '81; Macpherson et al., '82; Martino and Strick, '87; Hutchins et al., '88). Because the present materials do not distinguish between the known physiological fields within Region A, no inference regarding the number of physiological fields within the other regions nor the total number of fields can be safely made. Indeed, the very small "valley" in the labeling density between the primary sensory and primary motor cortex in the Old World monkeys described here was found only because it was painstakingly sought—and yet the separation of the two corresponding physiological fields has been routinely demonstrated in the same animals with electrophysiological methods.

Therefore, the present data do not mitigate nor distract from the functional significance of the multiple representations of the body musculature already shown to be present by other techniques (e.g., Strick and Preston, '82). It is only

the likelihood that the *number* and size of discrete origins of the CST and the number of physiological representations of the body musculature may also vary among mammals while the cortical *locations* of regions originating CST fibers are qualitatively stable which may prove to be a conclusion suggesting further experimentation.

Predominant ipsilaterality of the CST in some animals

The predominant ipsilaterality of the CST in the European hedgehog (*Erinaceus europus*) has been known for sometime. Verhaart ('70), using orthograde myelin degeneration methods, and Campbell ('65), using orthograde fiber degeneration methods, showed that the few fibers of the CST in European hedgehog (*E. europus*) penetrated the spinal cord only a segment or two without sign of a significant decussation. Because Verhaart ('70) also observed these same characteristics in mole, the only other Insectivora investigated, it was natural that both he and Campbell should suggest that the order of Insectivora might have only a rudimentary CST which was predominantly ipsilateral in termination.

In our original sample of mammals, we included only 1 species of hedgehog (*E. europus*) and 1 species of mole, and then only to be sure that some mammals with atypical spinal cord projections were represented. We wished to see if the cortical origins of their CST might also differ from other mammals.

Because we, too, had prepared hedgehog materials with Fink-Heimer ('67) and Ebbesson-Heimer (Ebbesson, '70) techniques and had come to essentially the same conclusions as Verhaart ('70) and Campbell ('65), we were greatly surprised when the first hedgehog prepared by our present retrograde labeling technique showed more than 3,000 labeled cells in its contralateral hemisphere. After verifying this result in a second case of *E. europus* with particularly careful verification that the extracellular HRP had not included the dorsal column nuclei, we added representatives of 2 more varieties of hedgehogs—one from the same family (Erinaceidae) but different genus (*Hemiechinus*) and the other from the same genus (*Erinaceus*) but different species (*albiventris*). Although these cases showed lesser numbers of labeled cortical cells, they verified the predominant ipsilaterality of the hedgehog CST.

The moles (*Talpa*) also verified Verhaart ('70) by showing a predominantly ipsilateral CST projection. However, the mole material also showed many more labeled cortical neurons than would be expected on the basis of previous orthograde degeneration methods (Verhaart, '70). Furthermore, the mole material showed that the *density* of labeled neurons per unit area both in Region A and in Region B was the highest of all mammals in the sample. At this stage, therefore, we were reluctantly forced to conclude that even the most sensitive of then-current orthograde degeneration methods, though adequate for determining many important characteristics of the system, was not adequate for the estimation of the *number* of CST fibers.

The variation from our expectations of the sheer number of CST cells prompted us to add 2 more species to the sample. First, we added the rock hyrax (*Procavia capensis*) which Verhaart ('67) had shown to have a predominantly ipsilateral, but less rudimentary, tract than hedgehog.

The second addition to the sample was selected on the basis of our observation that each of the animals with predominantly ipsilateral tracts, i.e., the 3 hedgehogs, the

mole, and now the hyrax, also had in common relatively small tracts and degenerate forepaws—forepaws with partially fused digits. Paleontology and comparative anatomy agree that the *primitive* condition of the mammalian forepaw is pentadactyly—5 separate digits. This means that any mammal deviating from pentadactyly either with apparently extra digits (*Panda*) or with fused digits such as Artiodactyla (antelope, cattle, sheep, pigs, etc.) or Perisodactyls (horses, tapirs, rhinoceros, etc.); or with partly fused digits such as hedgehogs or moles, are "deviant" in the comparative sense of that term.

Therefore, we sought an addition to our sample which was a legitimate member of the order Insectivora (as hedgehog and mole) but at the same time had 5 clearly unfused digits, true pentadactyly. We were able to satisfy these conditions by acquiring several individuals of the species *C. parva*—the least shrew. In addition to satisfying our 2 necessary conditions of being a member of the order Insectivora with a true 5-digit forepaw, the family of shrews (Soricidae) also has the distinction of being the modern animal most often referred to by paleontologists when describing "prototypical" or ancient placental mammals (e.g., LeGros Clark, '59; Walker, '64).

As already described in the previous section and shown in Figure 8, the cortical origin of the CST in the shrew was *not* predominantly ipsilateral as in hedgehog or mole (or hyrax). Instead, the ratio of contralateral to ipsilateral labeling was closely similar to the 17 other species; approximately 20 contralateral to 1 ipsilateral.

Because among Insectivora the family of shrews (Soricidae) has more genera (20) and more species (200) than any other, and because more of its morphological characteristics are unspecialized than any other member of the Insectivora, we are inclined to assign greater weight to the shrew's neurological characteristics too (e.g., Walker, '64). Therefore, we suggest that a predominantly ipsilateral CST probably should *not* be considered to be a characteristic of the order Insectivora despite its presence in hedgehogs and moles—both of which are representatives of much smaller and more specialized families.

Alternatively, we suggest that the predominantly ipsilateral CST of these latter animals might be a characteristic accompanying their fused digits. In any event, however, the relatively small but typically contralateral CST of the shrew and its similarity with the marsupials in the sample suggest that a predominantly contralateral CST is more likely the norm for mammals and also the more generalized and more primitive form, while a predominantly ipsilateral CST is the exception and possibly a degenerate form resulting from the secondary loss of pentadactyly along some lineages.

Uniformity of corticospinal origins

Although the presence of either 2 or 3 sites of CST origin has physiological implications, the commonality of the absolute and relative location of Regions A and B also has comparative and practical implications. Heretofore, with only fiber trajectory and spinal cord termination data available in sufficient amount to allow comparative and evolutionary conclusions regarding the CST, the critical differences in the form of the CST among mammals seemed to far outweigh the commonalities (e.g., Heffner and Master-ton, '75, '83). For example, the observation that the CST descends through the spinal cord in either the lateral, ventral, or dorsal funiculi in differing orders of mammals has been interpreted to mean that the CST probably arose

independently several times in the evolution of mammals—at least once for each funicular location, and possibly, once for each order of Mammalia (e.g., Goldby, '39; Noback, '59; Noback and Shriver, '66, '69).

Similarly, the differences in termination pattern of the CST fibers in the spinal cord gray have also been cited as grounds for suggesting the CST, strictly defined, may not be truly homologous among mammals but possibly a result of parallel or convergent evolution (e.g., Noback and Shriver, '66). In this case, it is the inconstant, presumably "premotor" terminations of the CST (i.e., in Rexed laminae 7, 8, and 9) as opposed to the relatively constant, presumably "descending sensory" terminations (i.e., in laminae 3, 4, 5, and 6) that vary widely among mammals (Heffner and Masterton, '75, '83).

Although this evolutionary argument regarding homology among CST tracts and hence their possibly multiple origin might appear to be of only paleontological interest, it has powerful practical implications. For example, if the CST of each order of mammals has no historical commonality—that is, if the CST did indeed evolve independently in each order of mammals—then useful animal "models" of the human CST are to be found only among man's own order, primates. In this case, information based on experiments in rats, cats, and even raccoon would have dubious applicability to clinical neurology due to the only "superficial" commonalities of the CST in these animals with the CST in humans.

For this reason at least, the striking similarity of Region A and Region B with their respective counterparts in non-primates is a tentatively reassuring observation. The likelihood that at least the 2 major subdivisions of the CST are analogous if not strictly homologous is markedly increased. And if the tract's differing termination patterns can be shown to be the result of selective pressures of a common kind (e.g., for digital dexterity or at least independent movement of the digits), the generalization of physiological observations even from rodents and carnivores to Man becomes a less risky proposal.

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